

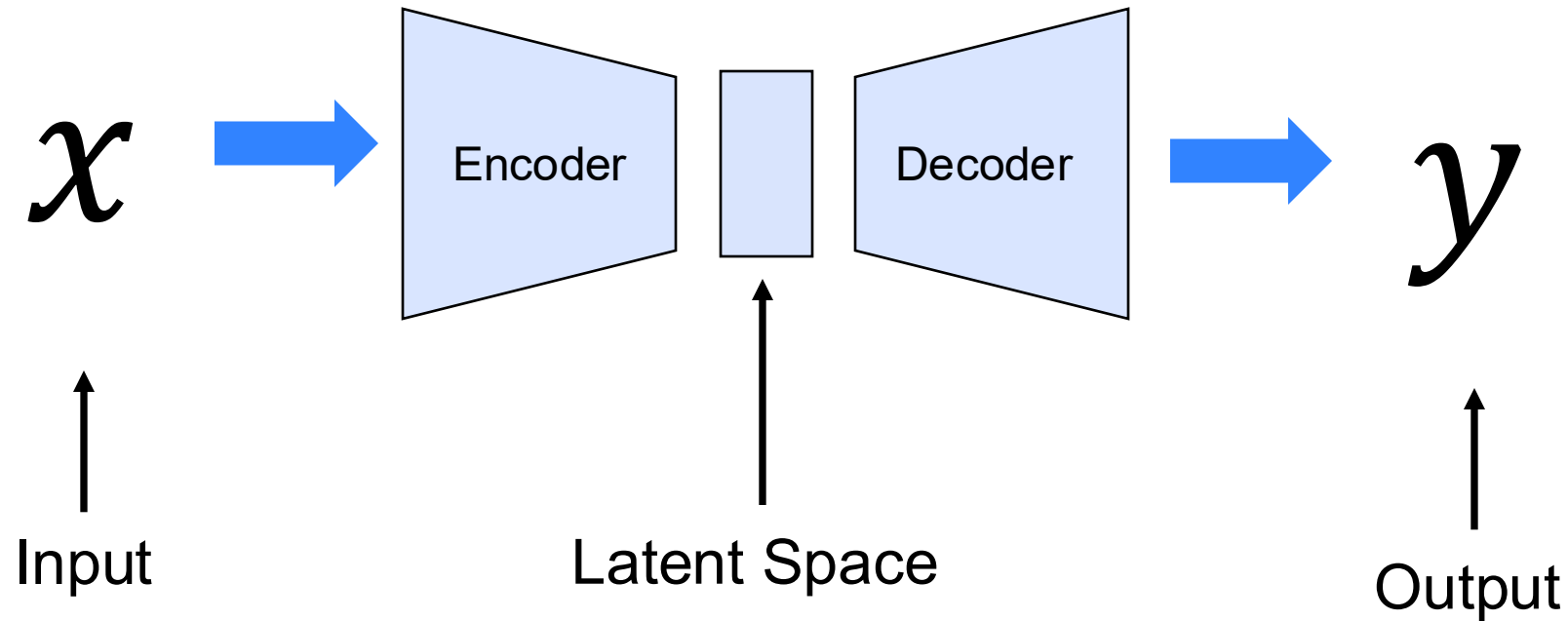


Artificial Intelligence in Interactive Systems

Encoder-Decoder Network, Autoencoder, and Variational Autoencoder

Encoder-Decoder Network

Goal: Compressing information into the latent space

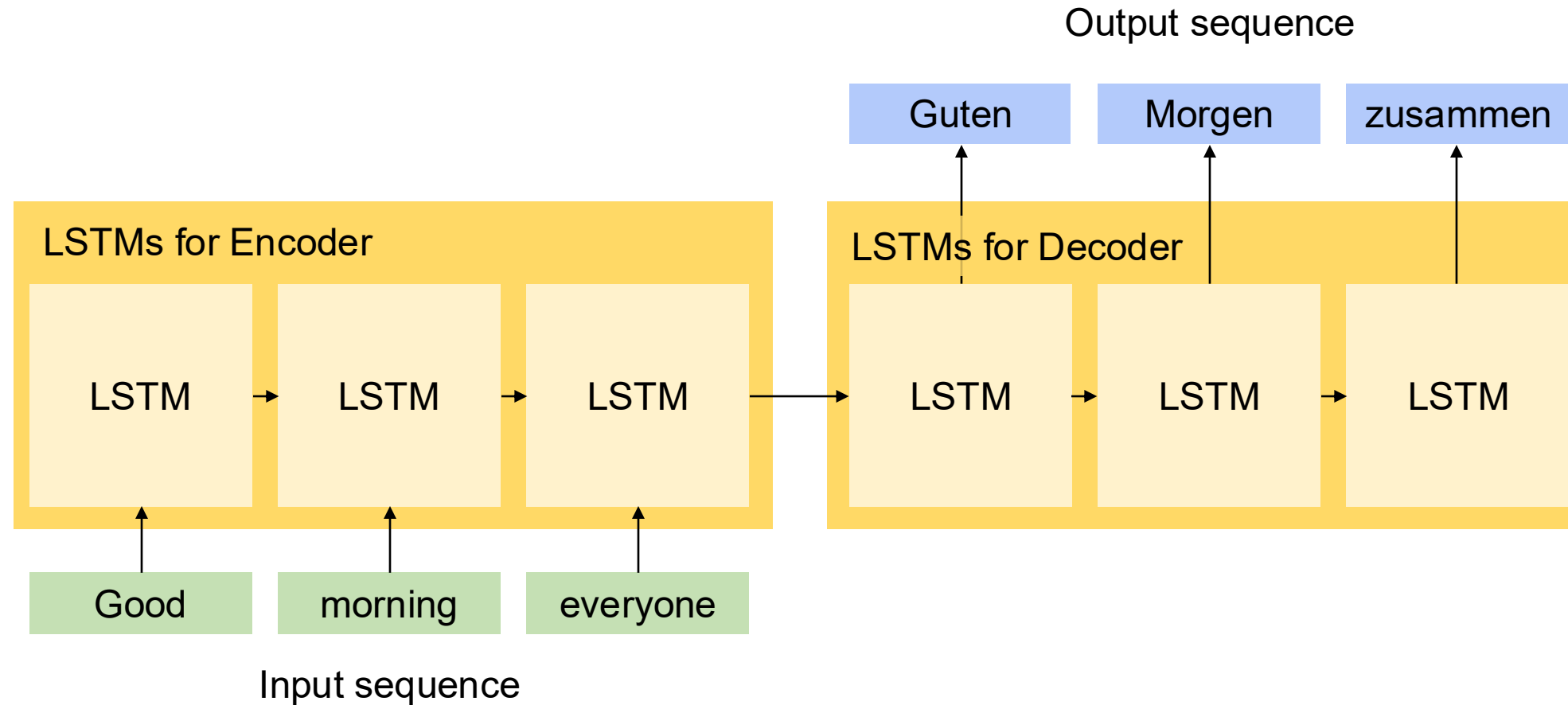


(Variational) Autoencoder

- Autoencoder
 - Mapping of one sample to one point in the latent space
 - Loss: reconstruction loss
- Variational Autoencoder
 - Mapping of one sample to a distribution “area” in the latent space
 - Loss: reconstruction loss + Kullback-Leibler (KL) divergence term

Sequence to Sequence Learning

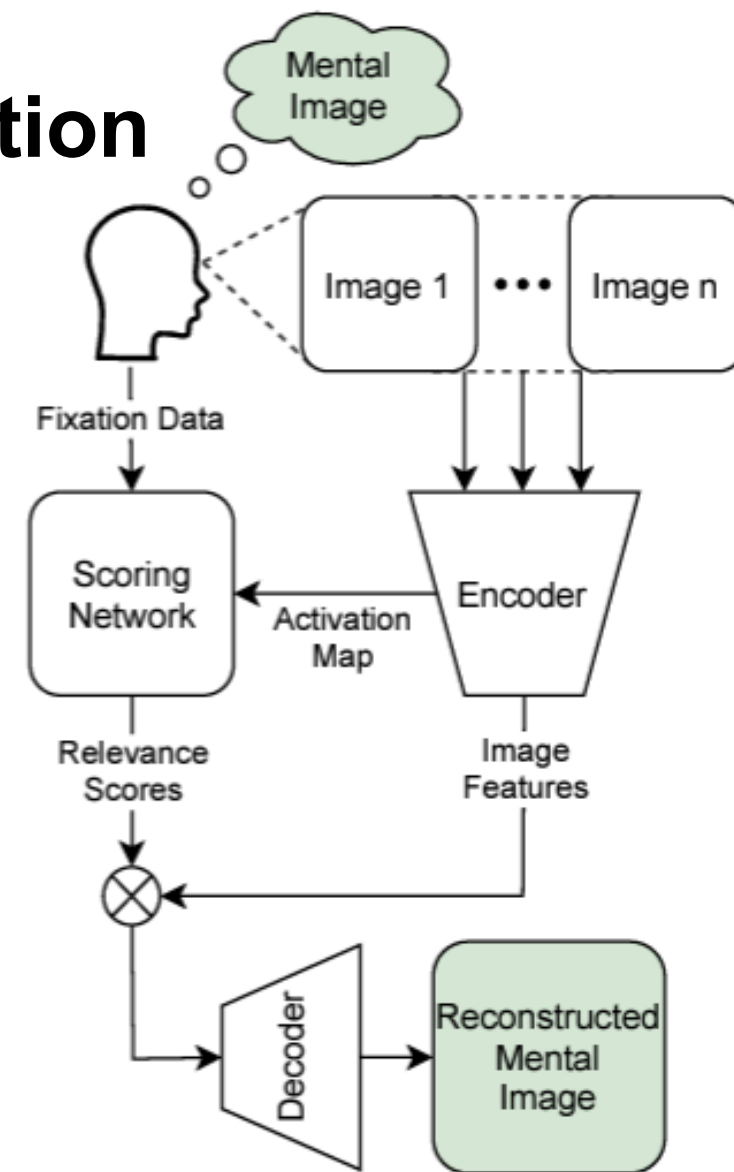
Language to Language translation



Sutskever, I., Vinyals, O., & Le, Q. V. (2014). Sequence to sequence learning with neural networks. *Advances in neural information processing systems*, 27.

Gaze-based Mental Image Reconstruction

An Example





Artificial Intelligence in Interactive Systems

Follow-up on Promoting

Zero Shot Prompting

- The model only gives the answer based on natural language that describes the task. Without successive improvements.

```
1 Translate English to French:
2 cheese => .....
```

← *task description*

← *prompt*

One-Shot Prompting

- In addition to the task description, the model also receives an example of the task. Without successive improvements.

1	Translate English to French:	← task description
2	sea otter => loutre de mer	← example
3	cheese =>	← prompt

.....

Few Shot Prompting

- In addition to the task description, the model is given several examples of the task. Without successive improvements.

The diagram illustrates a few-shot prompt for a translation task. It consists of five lines of text, each preceded by a line number (1-5). Line 1 is the task description: 'Translate English to French:'. Lines 2, 3, and 4 are examples: 'sea otter => loutre de mer', 'peppermint => menthe poivrée', and 'plush girafe => girafe peluche'. Line 5 is the prompt: 'cheese =>'. To the right of the text, there are labels with arrows pointing to the corresponding lines: 'task description' points to line 1, 'examples' points to lines 2, 3, and 4, and 'prompt' points to line 5.

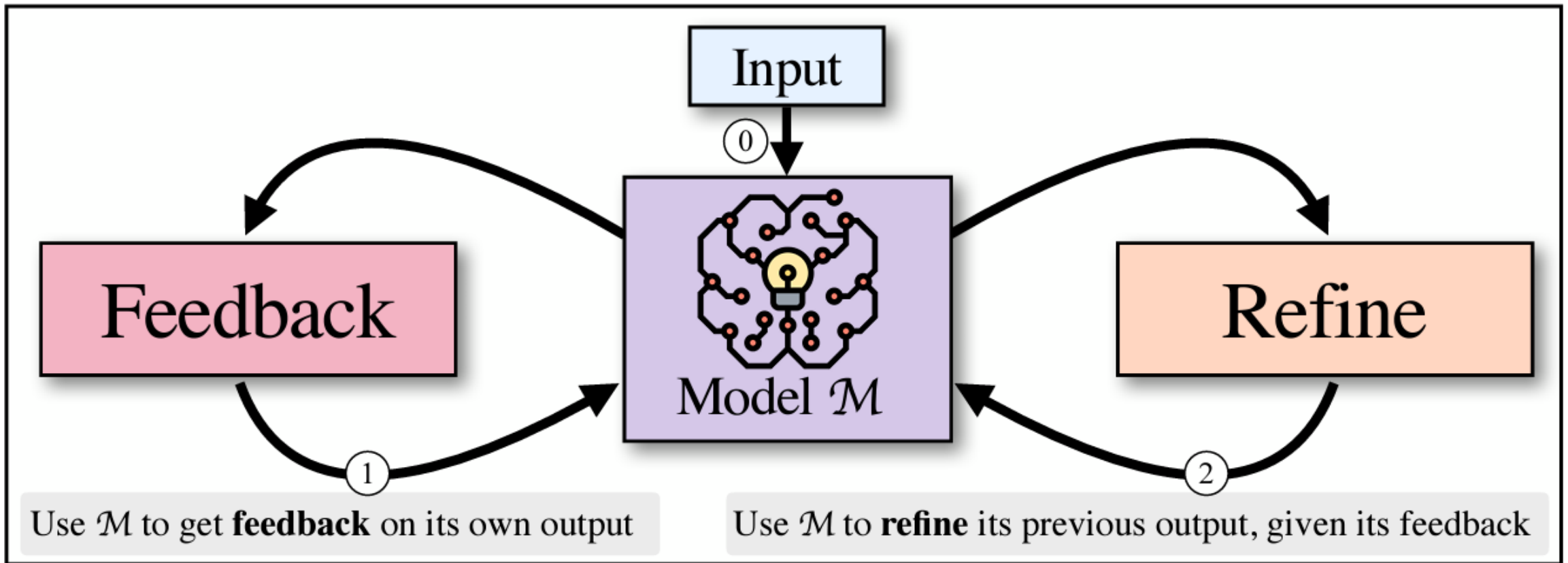
```
1 Translate English to French:
2 sea otter => loutre de mer
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese => .....
```

task description

examples

prompt

Self-Refine: Iterative Refinement with Self-Feedback



Self-Refine: Iterative Refinement with Self-Feedback

Task (e.g., create a person profile)

Prompt: What is a typical profile for a user who wants to do X?

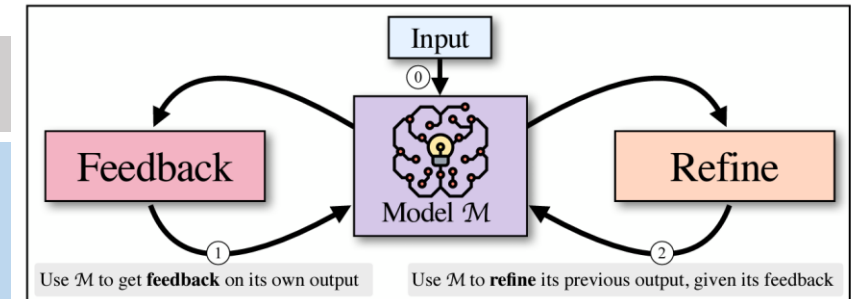
LLM answer: >Description of a profile<

Prompt: Give possible improvements to better describe the profile.

LLM answer: >Possible improvements<

Prompt: Improve the profile with the possible improvements

LLM answer: >Improved profile<



Divide and Conquer Iterative Creation

- *Prompt: >Description of a problem or task<*
- *Prompt: Divide the task into several subtasks*
- *Multiple prompts: Get answers to the subtasks*
- *...*
- *Prompt: Put the solutions together*

Prompt: I want to write an essay about X. What are the relevant aspects?

LLM-Answer: >List of Aspekten<

Prompt: What is a good structure for the essay?

LLM answer: >List of headings<

Prompt: What are keywords for heading Y?

> Keywords for Heading 2

-> Keywords for Heading 2

-> Schlüsselwörter für Überschrift 1

Prompt: Suggest a text for section Y?

> Text for Section 2

-> Text for Section 2

-> Text for Section 1

Prompt: Merge the text.

Chain-of-Thought Prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

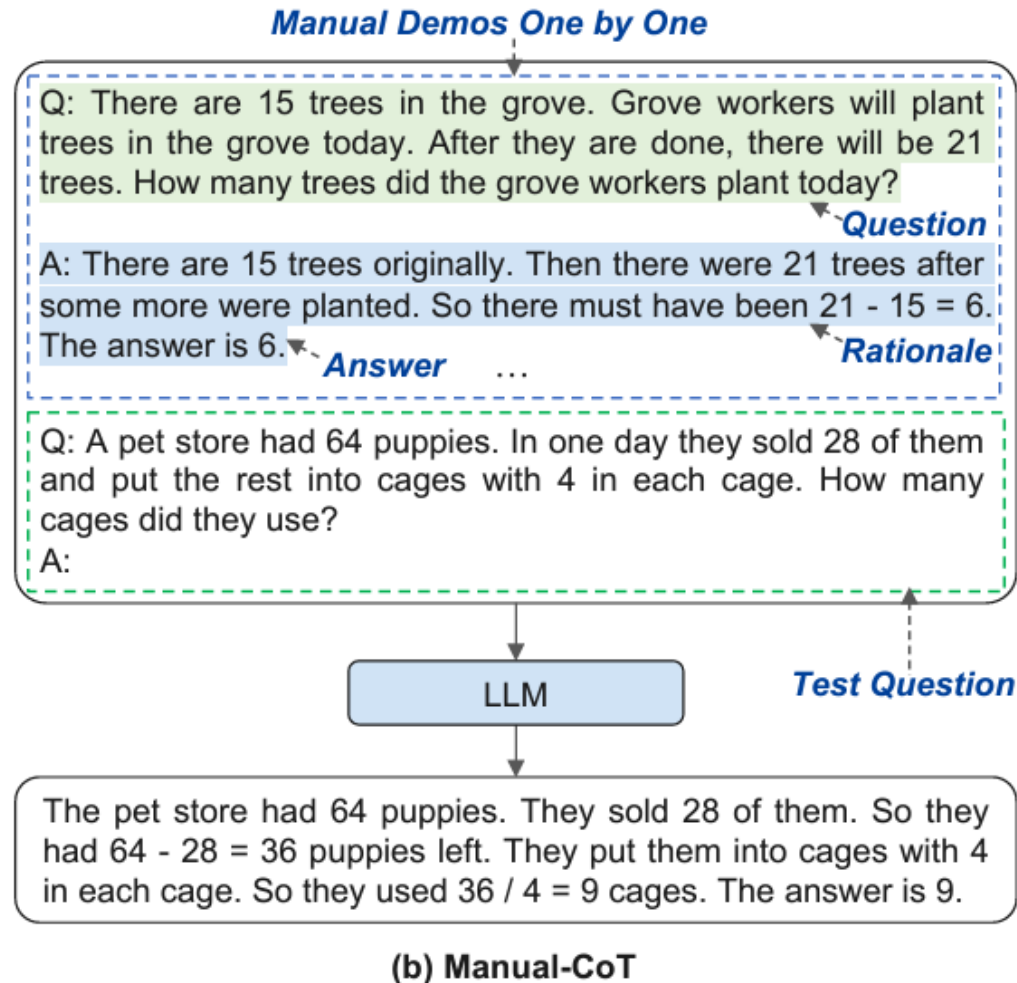
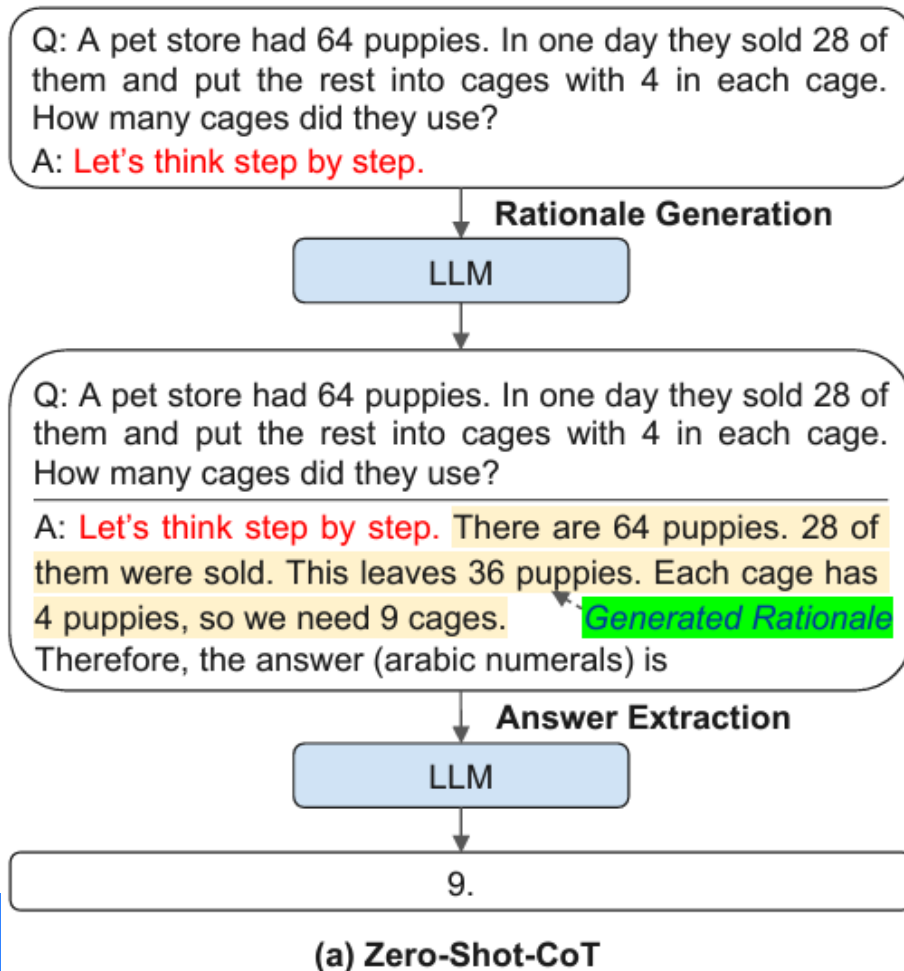
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Automatic Chain-of-Thought Prompting

“Let's think step by step”



Zero-shot COT Prompting

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. **X**

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. **✓**

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 **X**

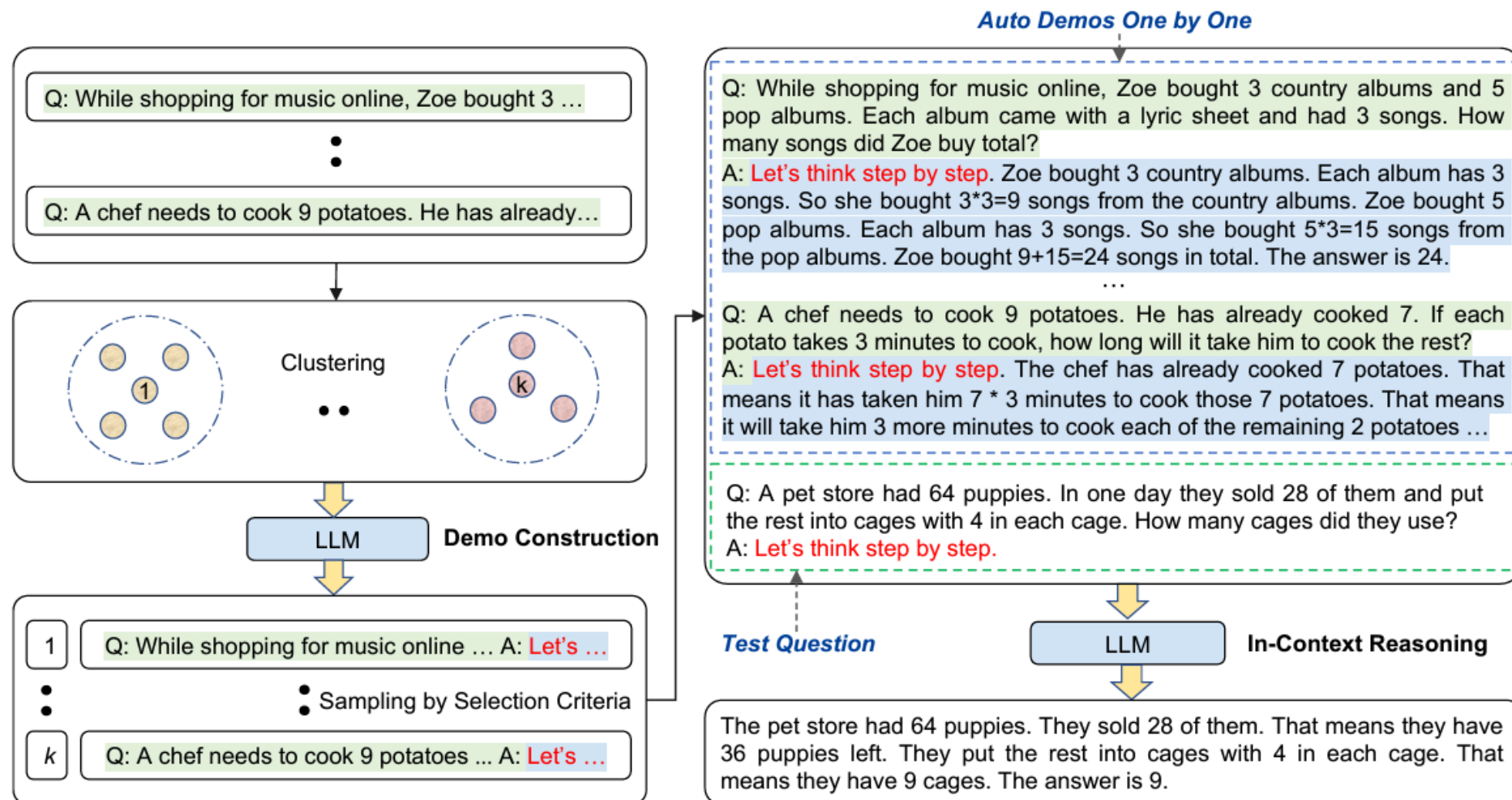
(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

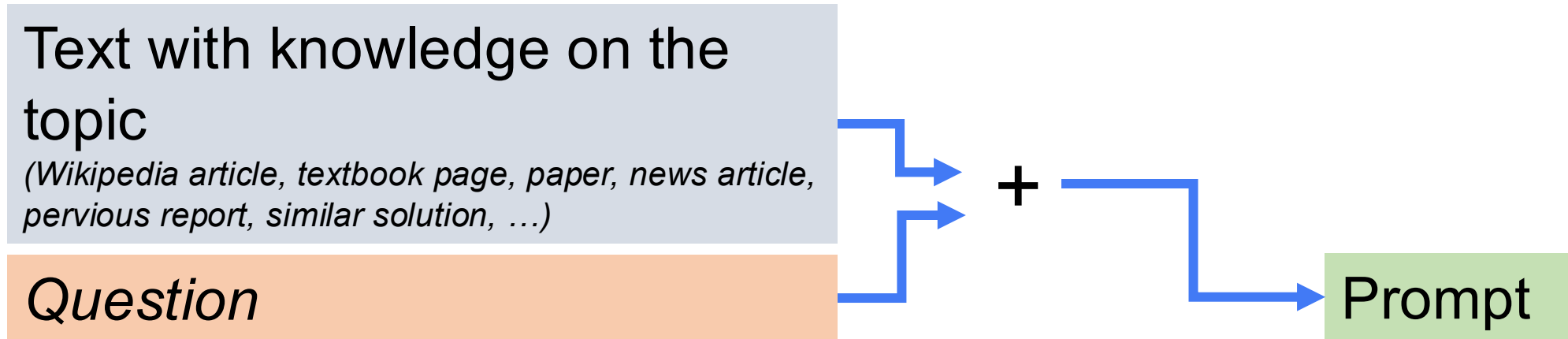
A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. **✓**

Automatic Chain-of-Thought Prompting

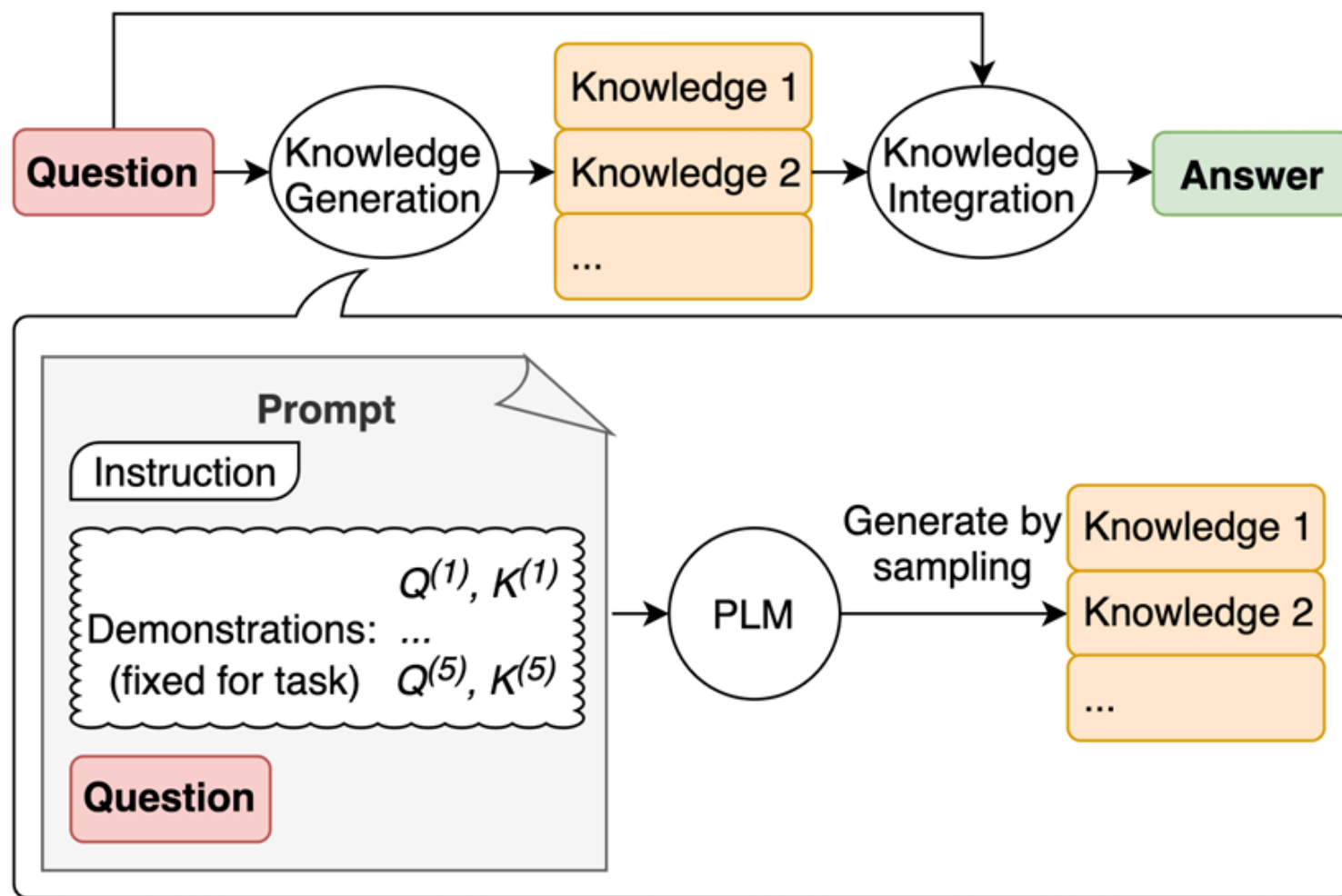


Prompting with Knowledge



- The knowledge contained in the prompt does not require any information to answer the question.
- The knowledge “guides” the answer in the right direction.

Knowledge Generation Prompting



Prompt Instruction Cheat Sheet

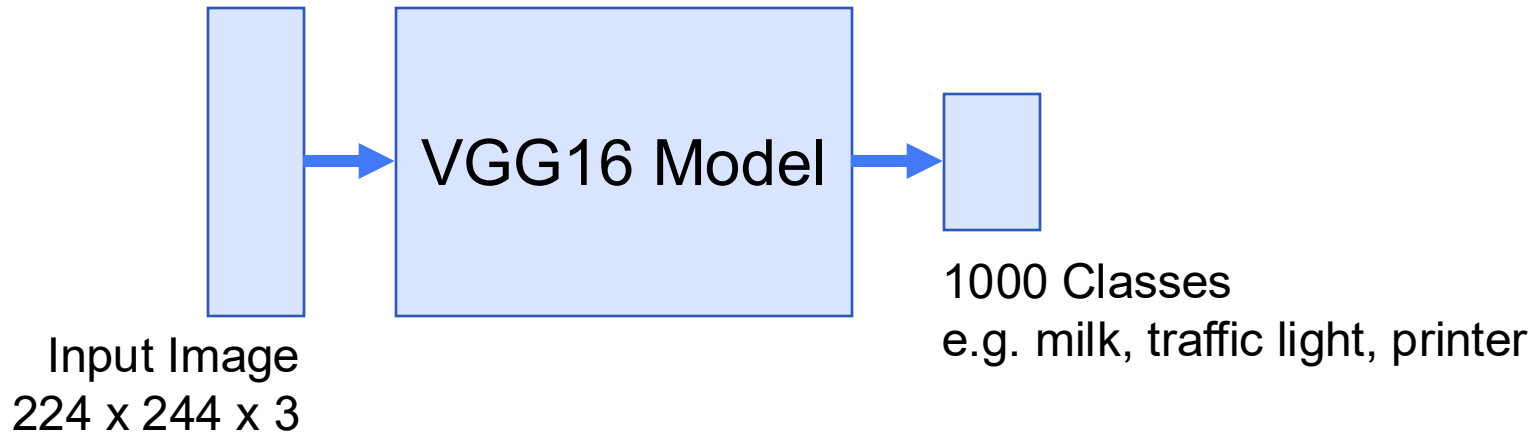
Content of the Instruction	Task of this Component	Dos (exemplary)	Don'ts (exemplary)
Bot Persona	Define a behavior with many individual facets in just a few words or for a story driven interaction with an NPC or avatar, define a bot persona with a backstory that makes the bot's behavior more exciting and interactive. Provide names, dates, places in the story.	Act as a sensitive elderly psychotherapist ... as a patient support staff ... as a professional journalist ... as a car in love with its driver ... you are Selma Brand, a 28-year old gunslinger who lost her family in a bandit raid on their farm. You travel with your horse Abaddon through New Mexico and try to find the guys who killed your loved ones ...	Act as a helpful AI
User persona	Define the level of detail, complexity, required knowledge of the recipient in a few words	Explain it like to someone with an MSc in software engineering ... like to a 5-year-old child	Answer the user ...
Verb	Define the task. To "answer" is not the task, because the model will always respond.	Summarize, list, ask, classify, explain, format, extract, translate to Spanish ...	Answer the question, give me, ...
Output format	Define the exact format of the output	A list of steps, a formula, python code, a JSON, a number, a recipe. [you can define the output format in a separate section as well]	Answer, text
Length, level of detail	Describe the output format and number of elements	Up to 3 sentences ... 10 examples with product name and price	Short, few, detailed
Tonality	Define the style, the way the model should convey the message	Empathetic, confident, aggressive, moaning, sarcastic, witty, stuttering	Friendly, neutral, smart, intelligent
Edge cases	Define responses for special situations: Model has no information ... does not understand the question ... if the user gets aggressive, toxic ...	If you can't answer the question based on the provided context, tell the user you can't ...	Answer only on the basis of your knowledge ...
Whitelist conversation topics	Define all the topics the model can talk about, not the other way round (list forbidden topics)	Discuss only the operation of the coffee machines of the manufacturer XYZ and talk about coffee culture. Friendly refuse to chat on something else.	Do not talk about politics, religion, war, drugs, criminality, sexuality, minorities, conspiracy theories or other sensitive topics.



Artificial Intelligence in Interactive Systems

Reinforcement Learning

Supervised Learning



- Labeled Data Needed
- The model can not be better than the labels

Learning Strategies

Supervised
Learning

Unsupervised
Learning

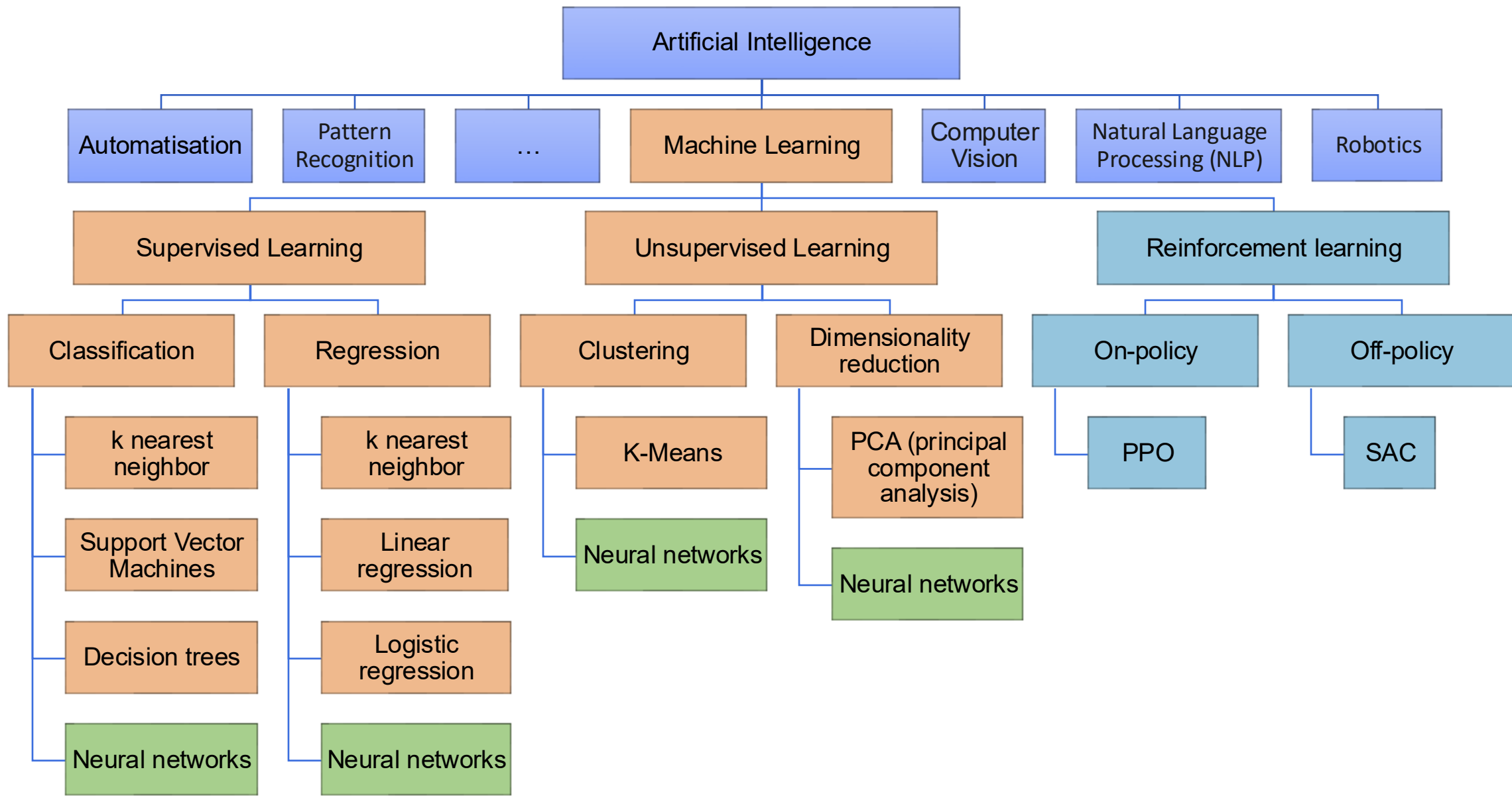
Reinforcement
learning

Learning Strategies

Supervised
Learning

Unsupervised
Learning

Reinforcement
learning



AI triumphs against the world's top pro team in strategy game Dota 2

It's the first time an AI has beat a world champion e-sports team.

By Kelsey Piper | Apr 13, 2019, 6:30pm EDT

f t SHARE



OpenAI demoed their Dota bot with live matches at an event in San Francisco. | Kelsey Piper/Vox

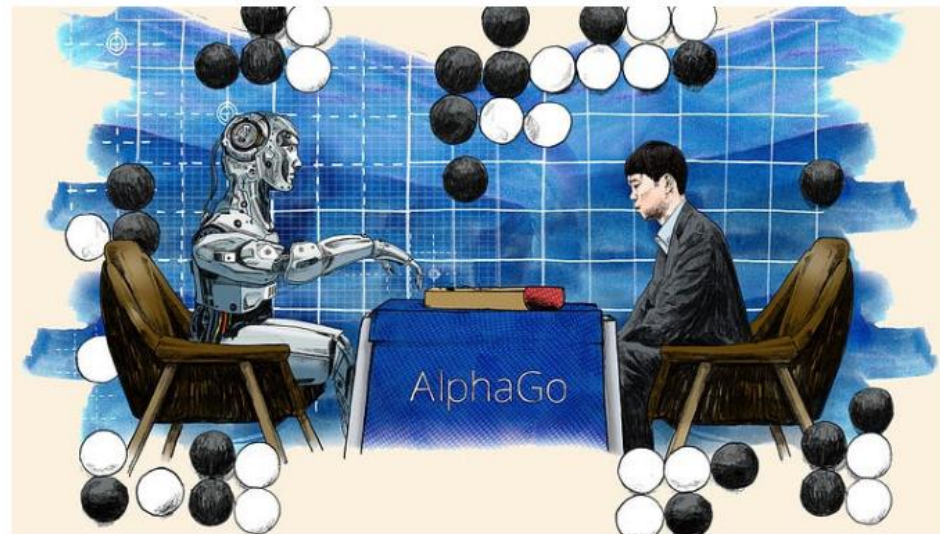
<https://www.vox.com/2019/4/13/18309418/open-ai-dota-triumph-og>

AlphaGo Beats The World's Best Go Player



Ashley · Follow

Published in Level Up Coding · 5 min read · Nov 16, 2020

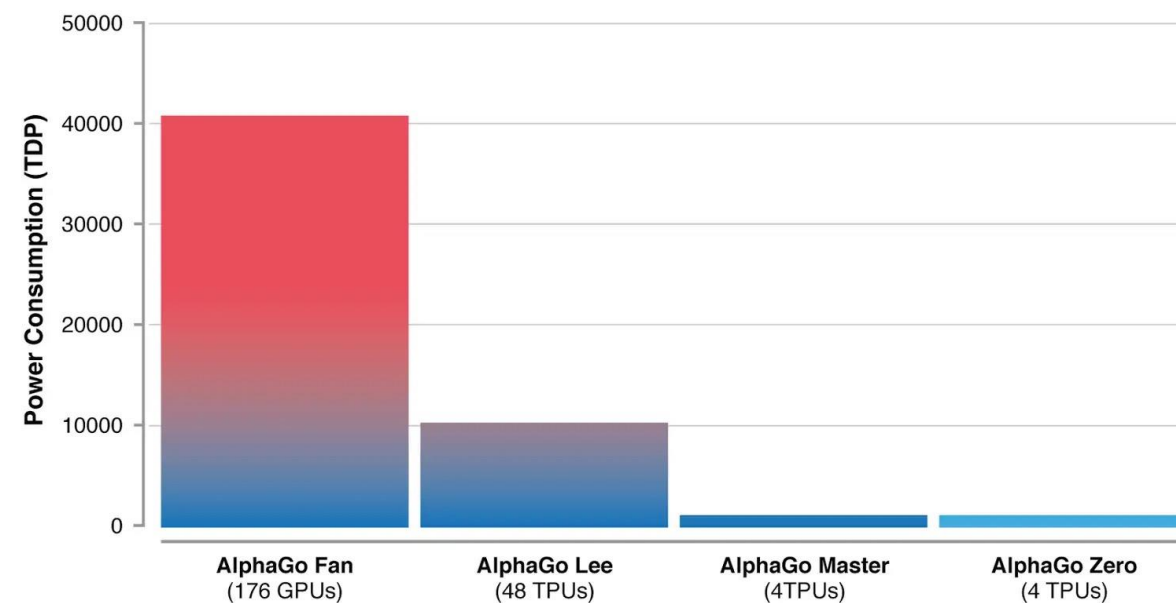
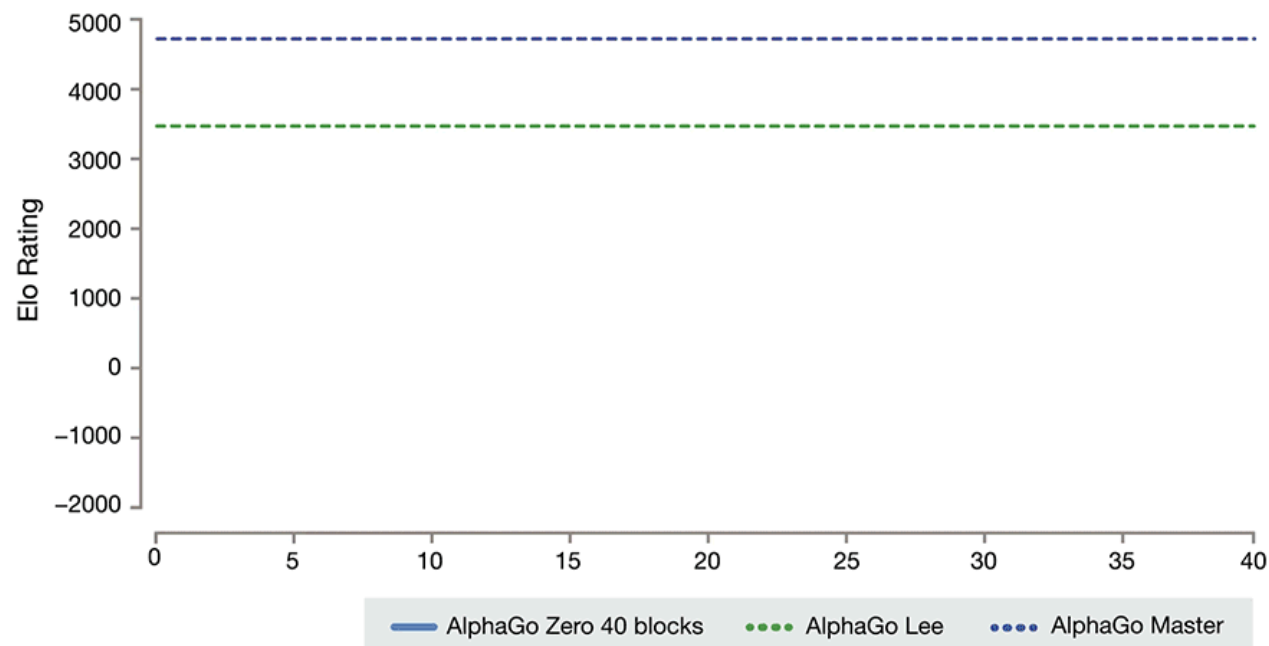


Not only is AlphaGo the first machine to defeat a professional human Go player, but the machine is also the first to defeat a Go world-renowned champion, and will always arguably be the strongest Go player in history.

<https://levelup.gitconnected.com/alphago-beats-the-worlds-best-go-player-1d4ab1428bac>

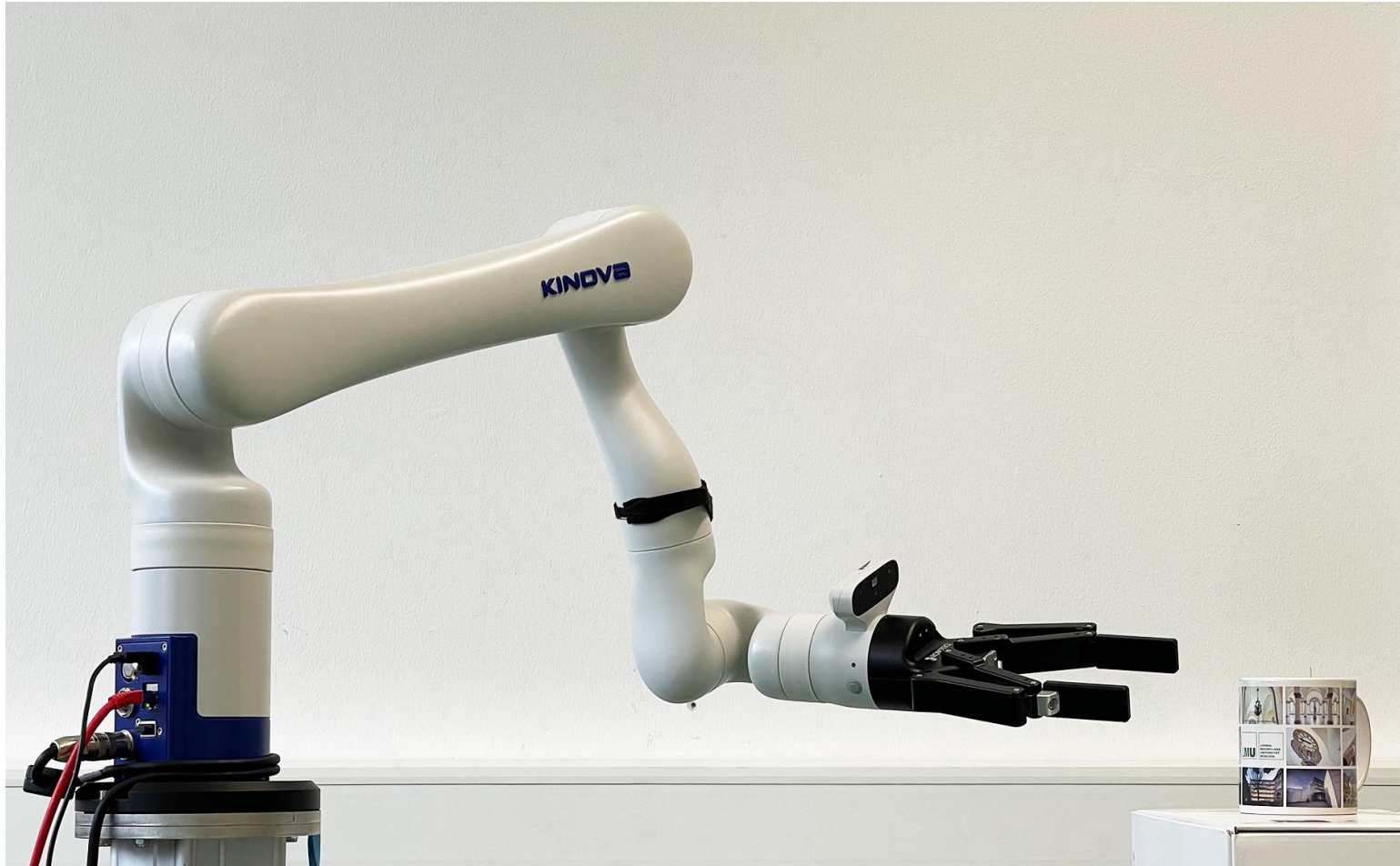
Do we need human observation?

Example AlphaGo Zero



<https://www.deepmind.com/blog/alphago-zero-starting-from-scratch>

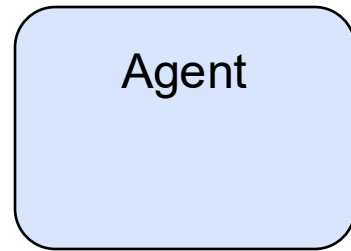
Pick Up the Cup



Reinforcement Learning

*Learning from the environment what works
by getting positive feedback/reward*

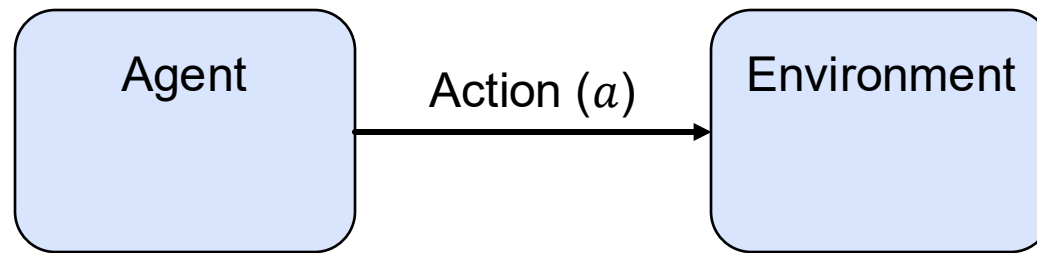
Reinforcement Learning



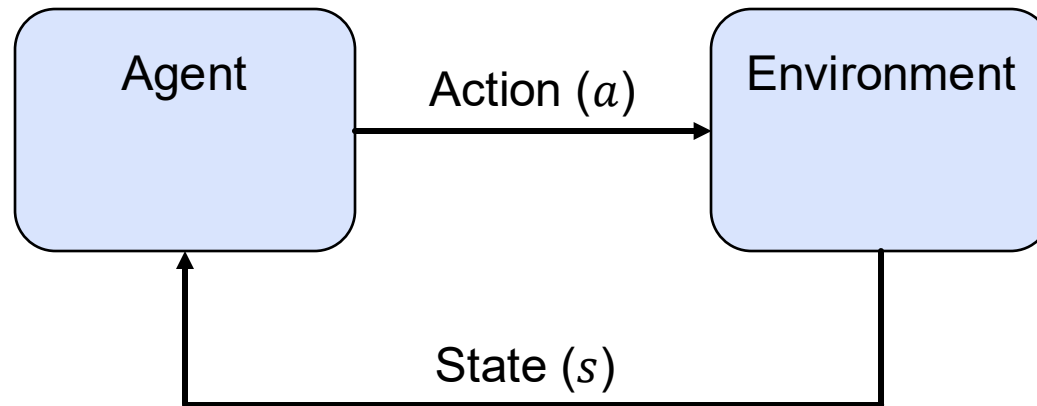
Reinforcement Learning



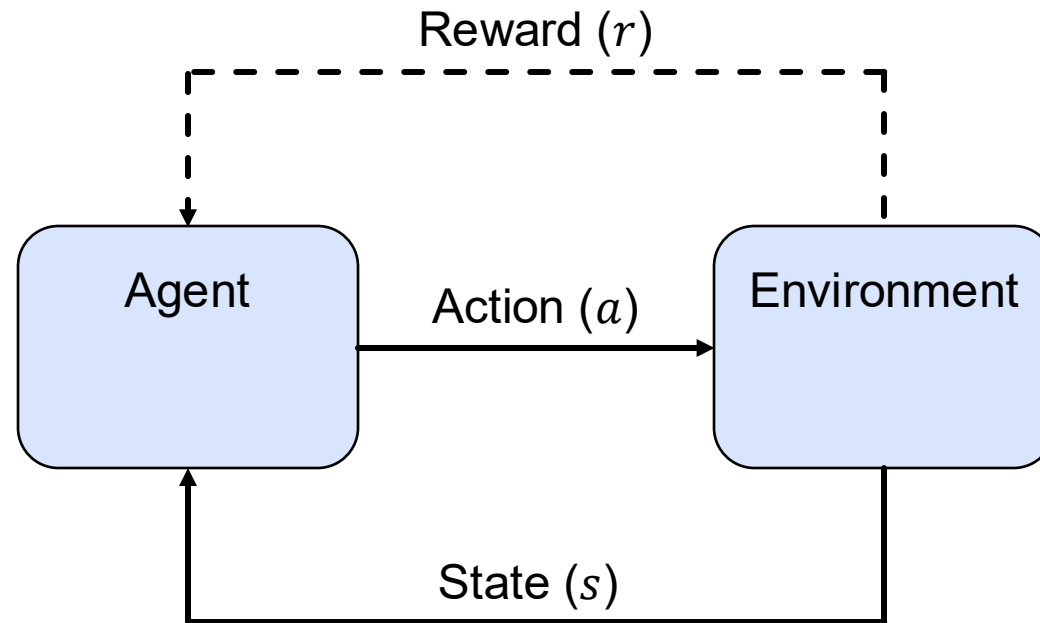
Reinforcement Learning



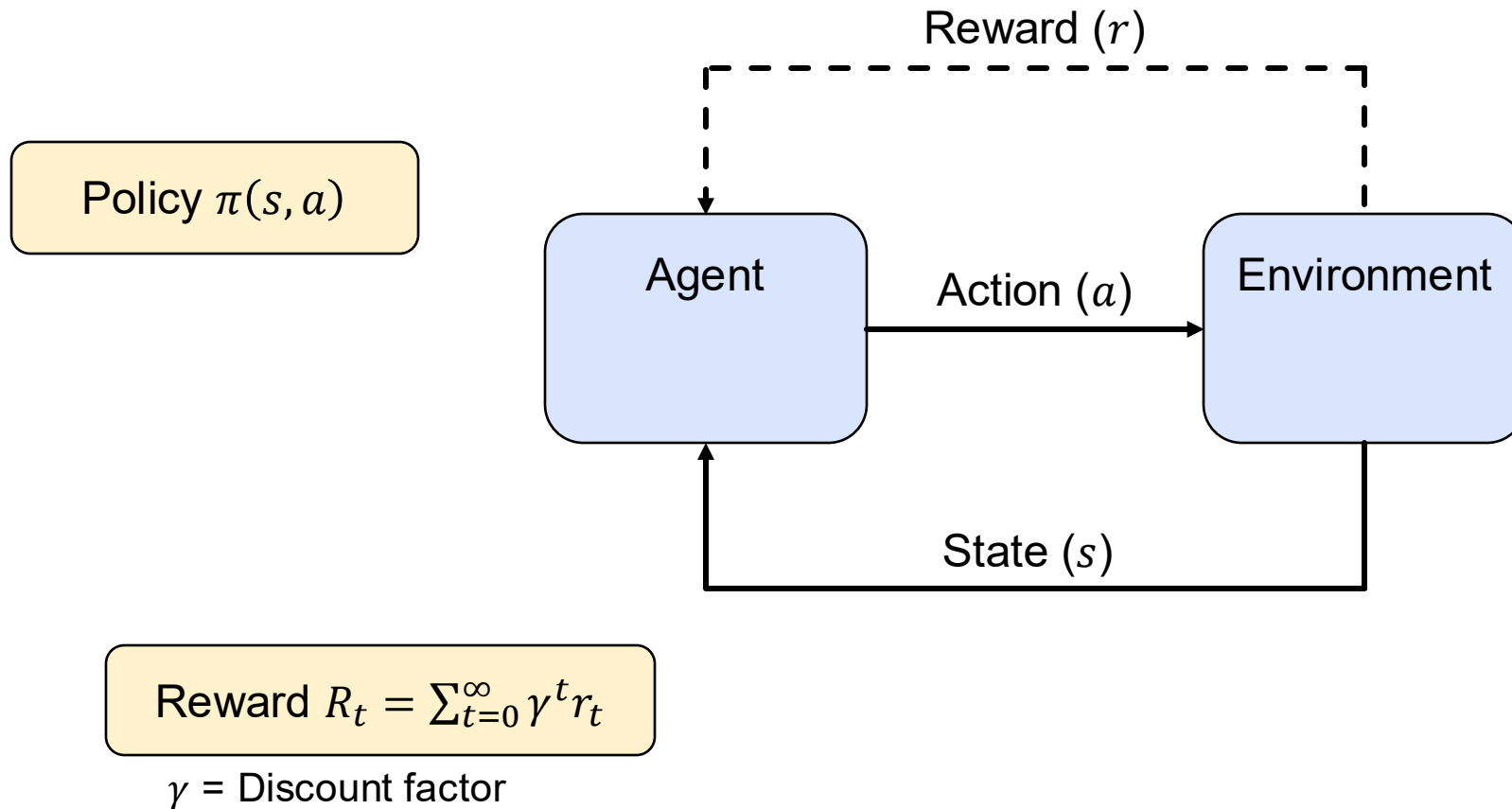
Reinforcement Learning



Reinforcement Learning

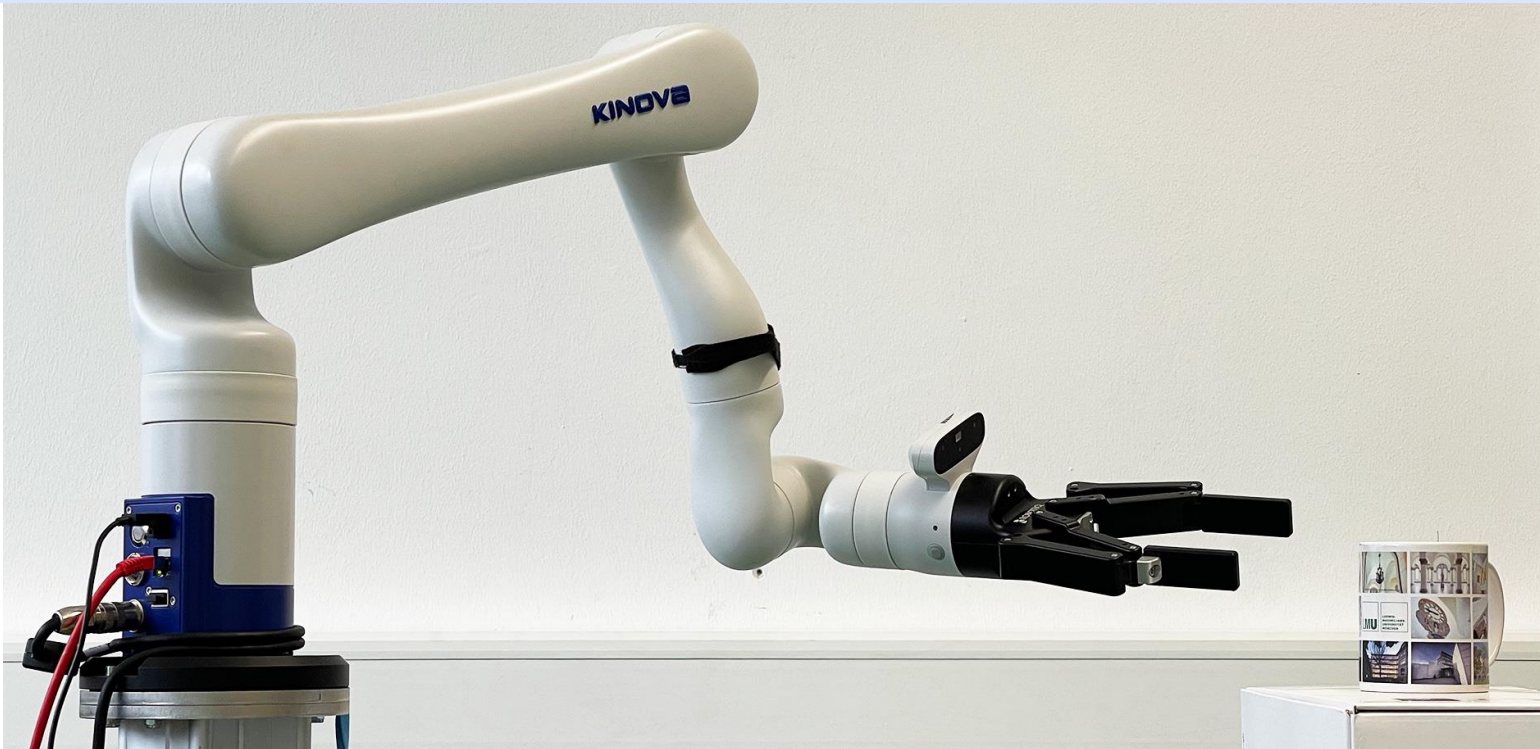


Reinforcement Learning



Credit Assignment Problem

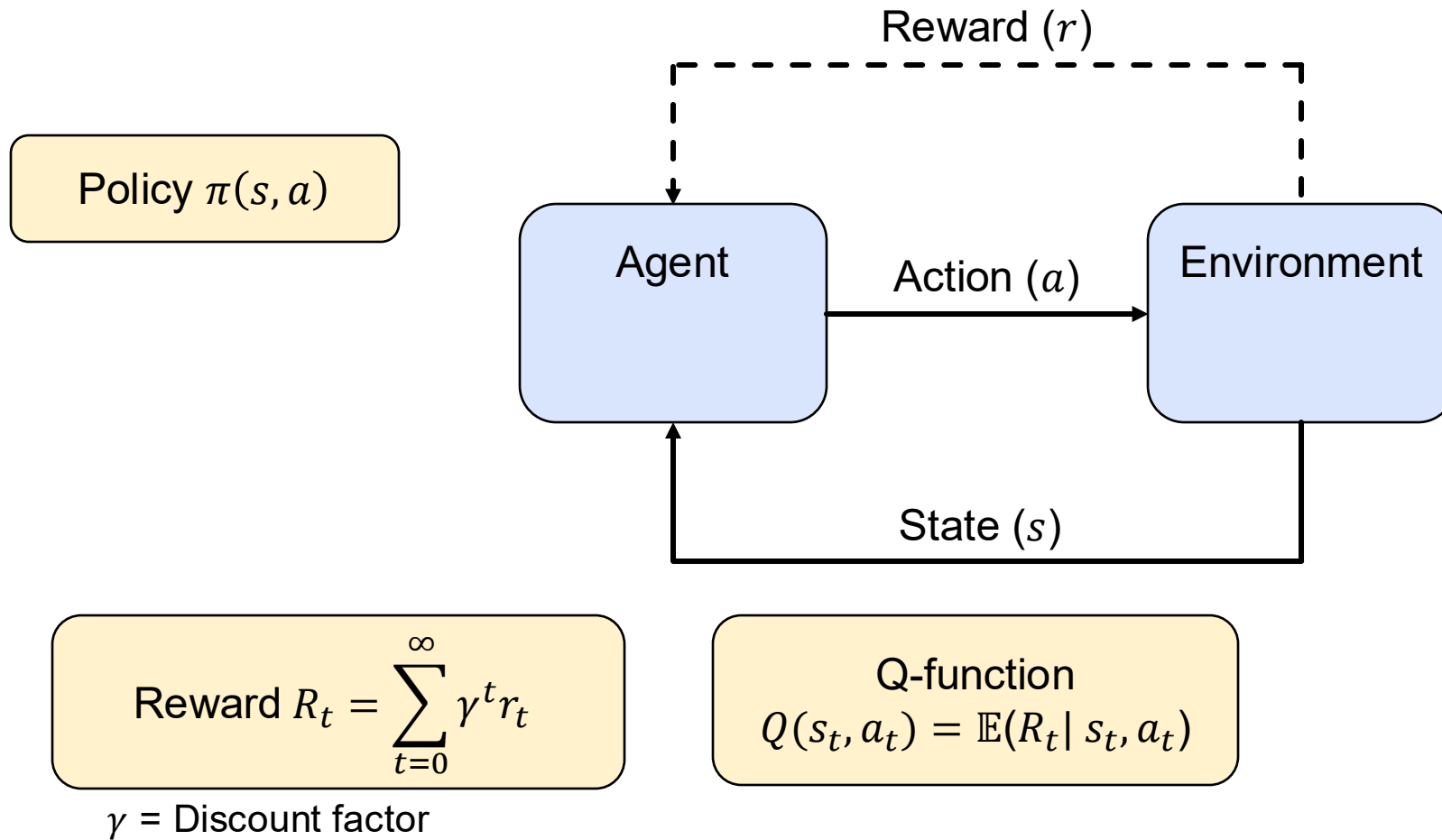
Which of the action contributed to getting the reward?



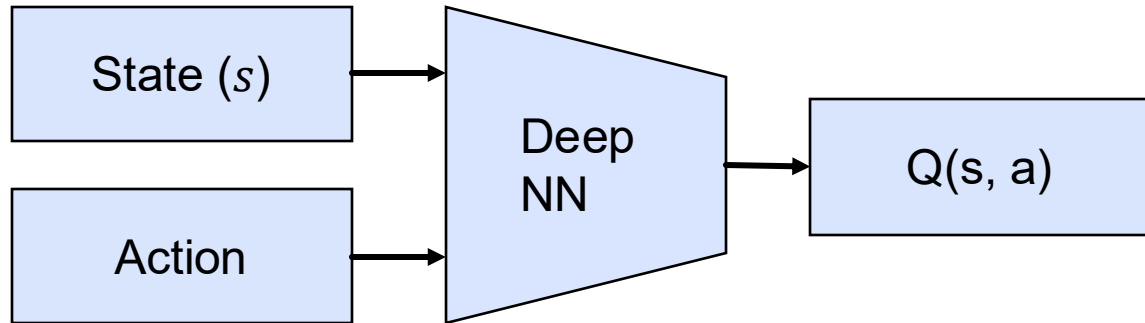
The credit assignment problem refers to the problem of measuring the impact on our environment with a given state (s) of an action (a) performed by an agent on future rewards (r).

Minsky, M. (1961). Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1), 8-30.

Q – Learning



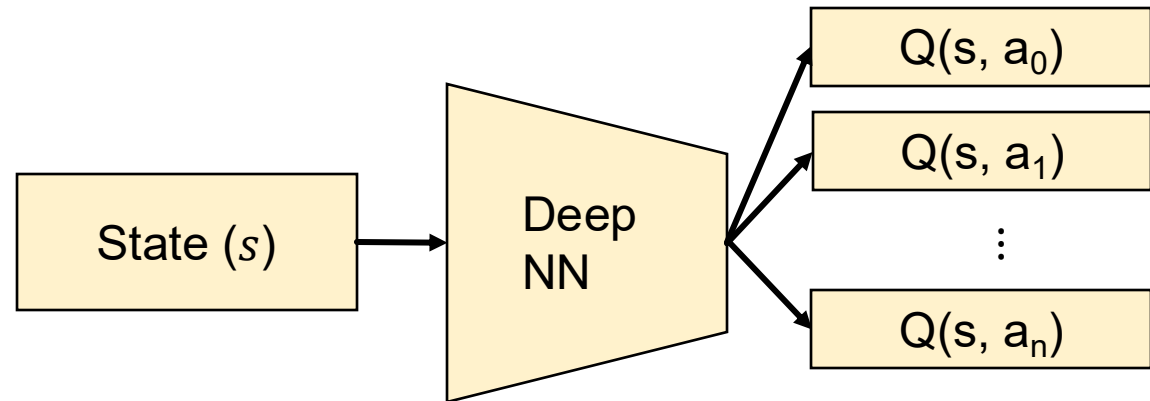
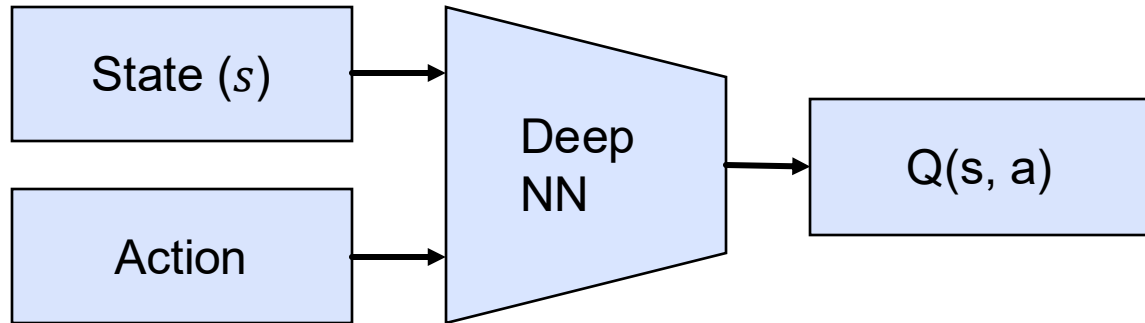
Deep Q Networks (DQN)



The goal is to approximate the Q-function

The model will predict the reward giving a state (s) and an action (a).

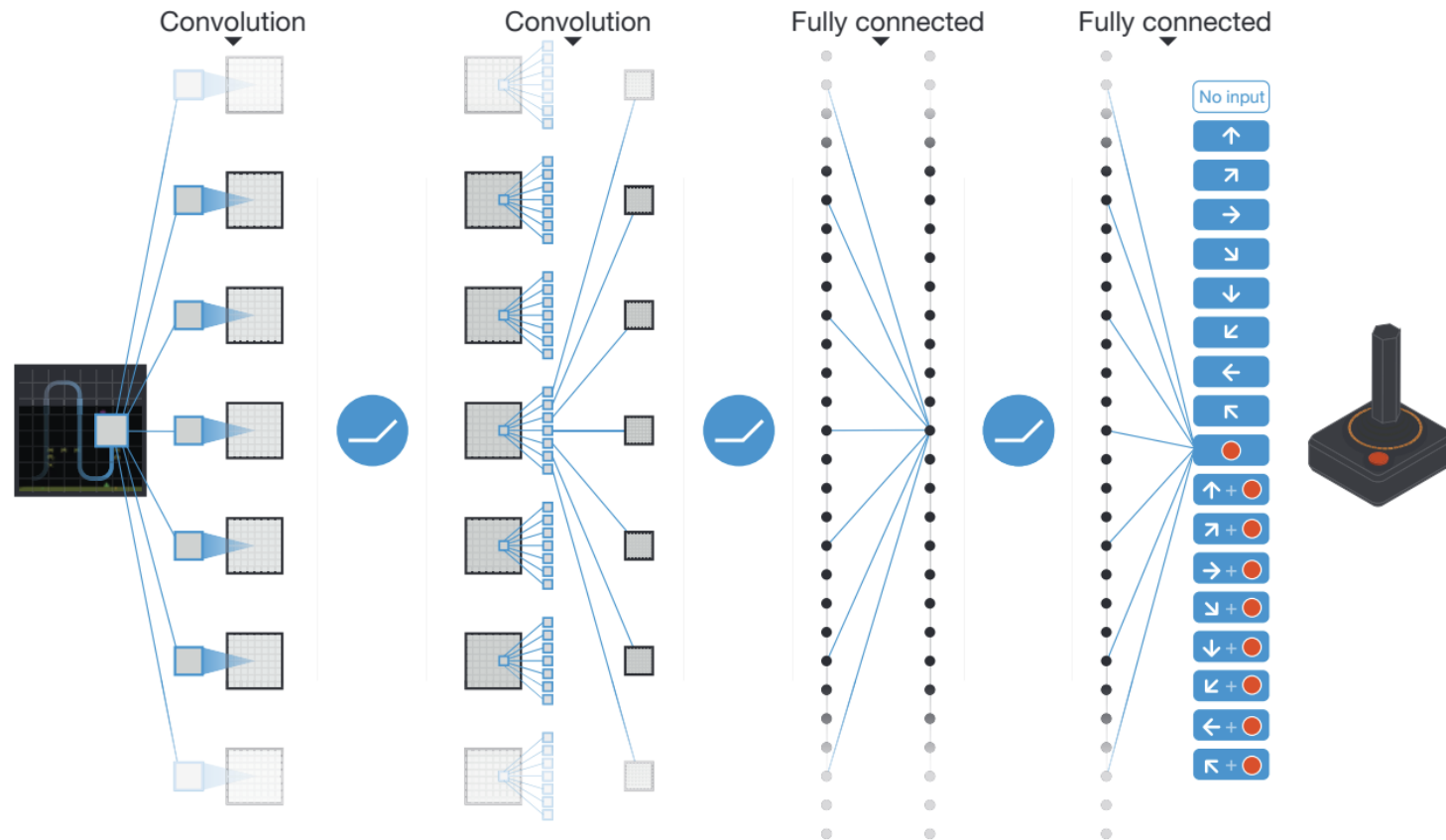
Deep Q Networks



Now we can get the rewards for all possible actions.

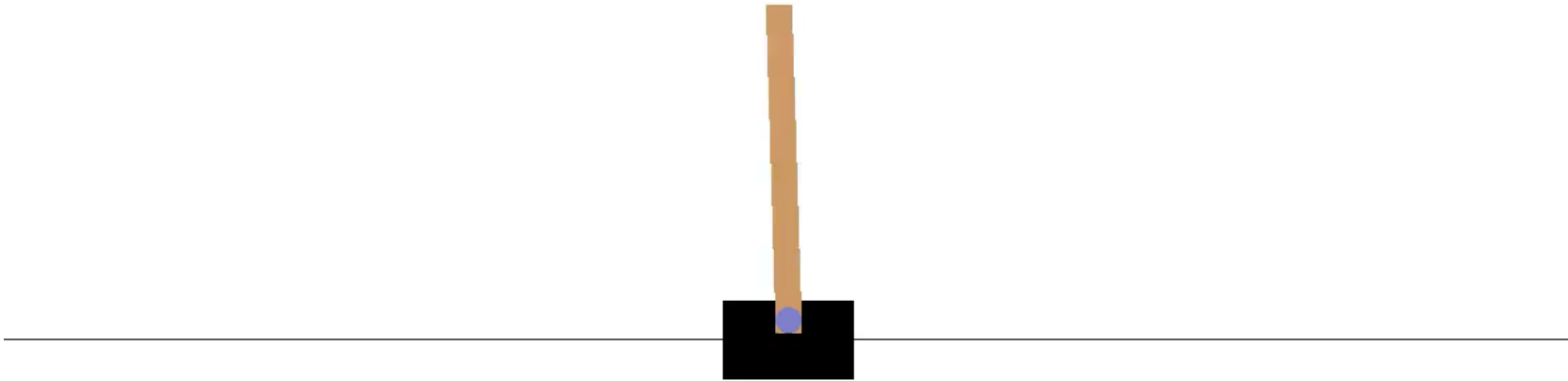
Mnih, Volodymyr, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves et al. "Human-level control through deep reinforcement learning." *nature* 518, no. 7540 (2015): 529-533. Credit: <https://www.youtube.com/watch?v=TmPfTptdgg>

Deep Reinforcement Learning



Mnih, Volodymyr, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves et al. "Human-level control through deep reinforcement learning." *nature* 518, no. 7540 (2015): 529-533.

Exercise



https://github.com/tensorflow/agents/blob/master/docs/tutorials/1_dqn_tutorial.ipynb

Conclusion

Reinforcement Learning

- What is Reinforcement Learning?
- Agent in environments
- Policy for next action
- Learning from reward

- Using Deep Neuronal Networks in RL
 - Deep Q Learning
 - Next steps: Policy Gradient
 - ... and many more approaches



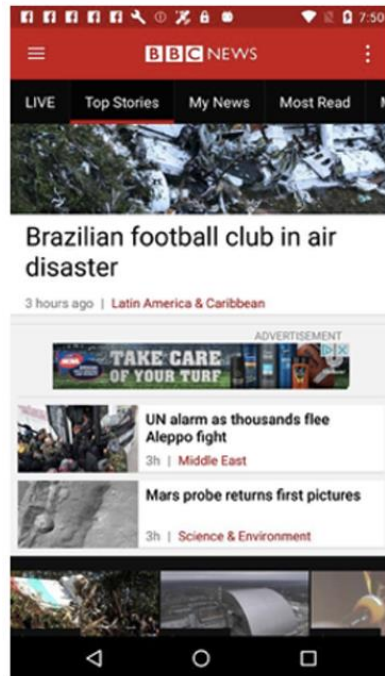
Artificial Intelligence in Interactive Systems

Applications

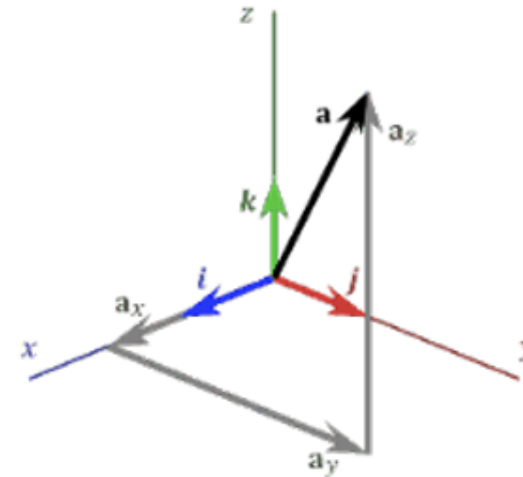
Screen2Vec: Semantic Embedding of GUI Screens and GUI Components

Click Cloneplot Master title style

GUI Screen Image



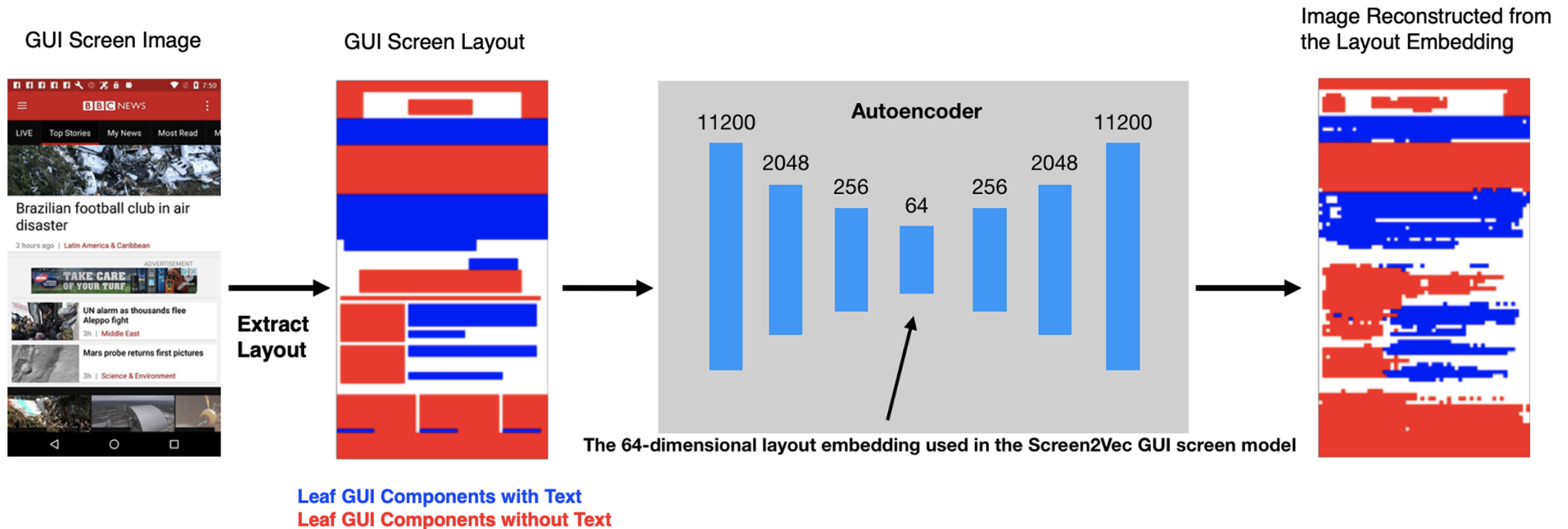
Output Vector



Toby Jia-Jun Li, Lindsay Popowski, Tom Mitchell, and Brad A Myers. 2021. Screen2Vec: Semantic Embedding of GUI Screens and GUI Components. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 578, 1–15. <https://doi.org/10.1145/3411764.3445049>

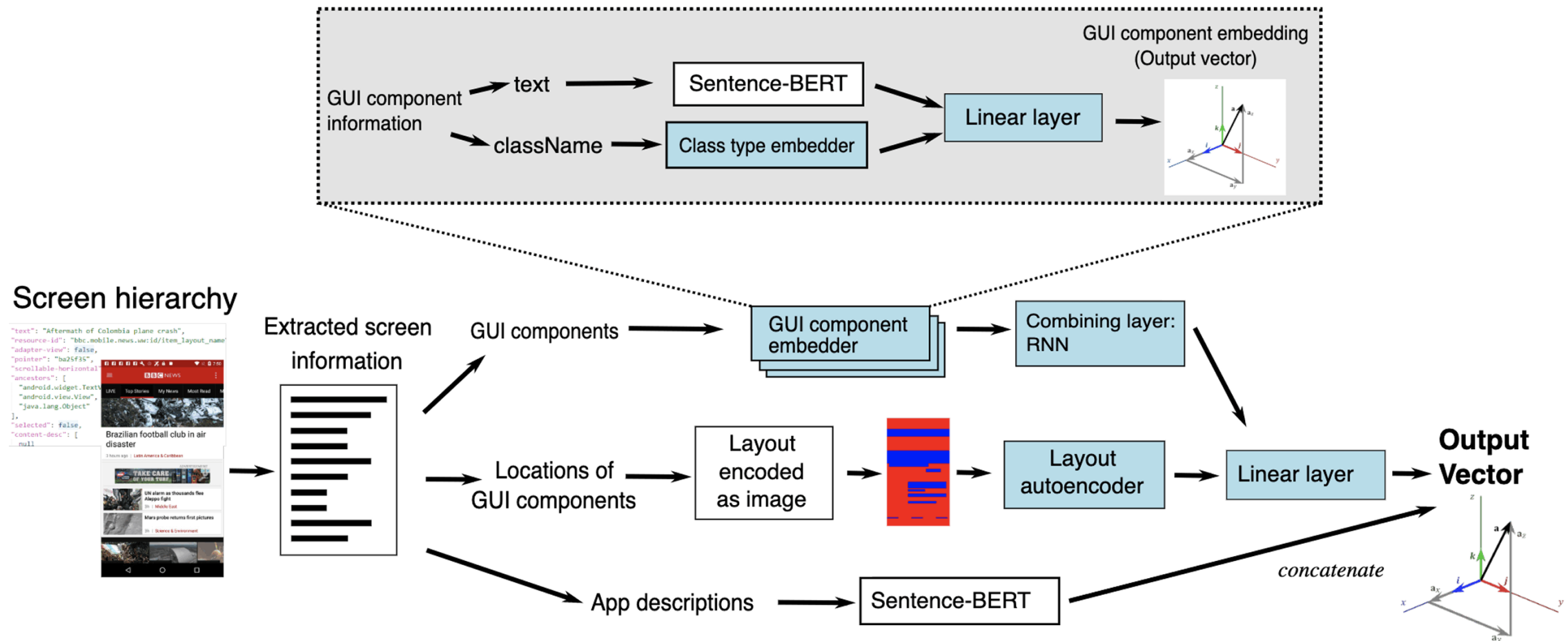
Screen2Vec: Semantic Embedding of GUI Screens and GUI Components

Glitter Clone in Matter title style



Toby Jia-Jun Li, Lindsay Popowski, Tom Mitchell, and Brad A Myers. 2021. Screen2Vec: Semantic Embedding of GUI Screens and GUI Components. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 578, 1–15. <https://doi.org/10.1145/3411764.3445049>

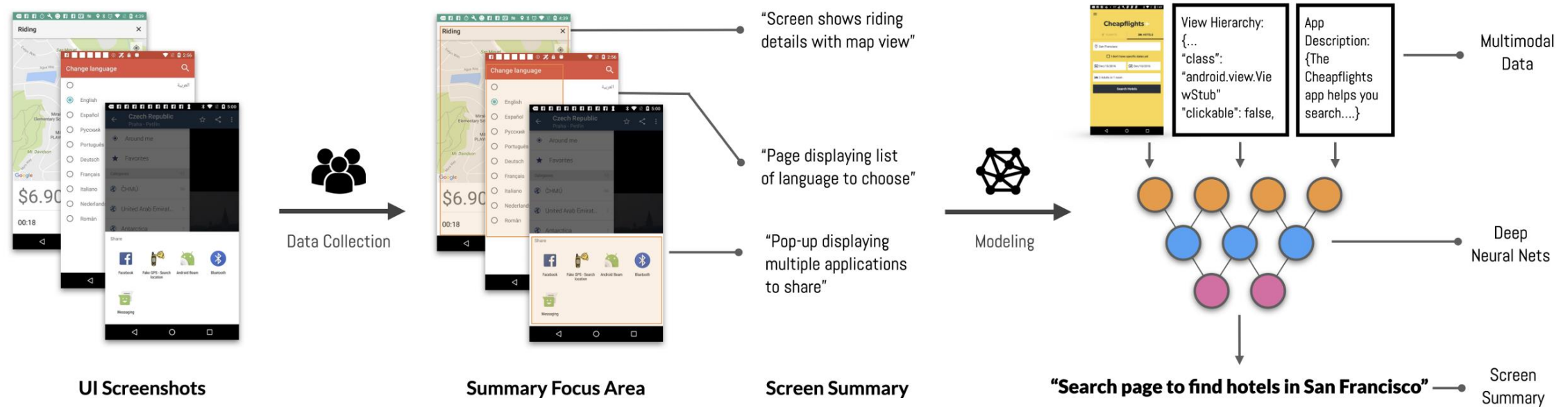
Screen2Vec: Semantic Embedding of GUI Screens and GUI Components



Toby Jia-Jun Li, Lindsay Popowski, Tom Mitchell, and Brad A Myers. 2021. Screen2Vec: Semantic Embedding of GUI Screens and GUI Components. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 578, 1–15. <https://doi.org/10.1145/3411764.3445049>

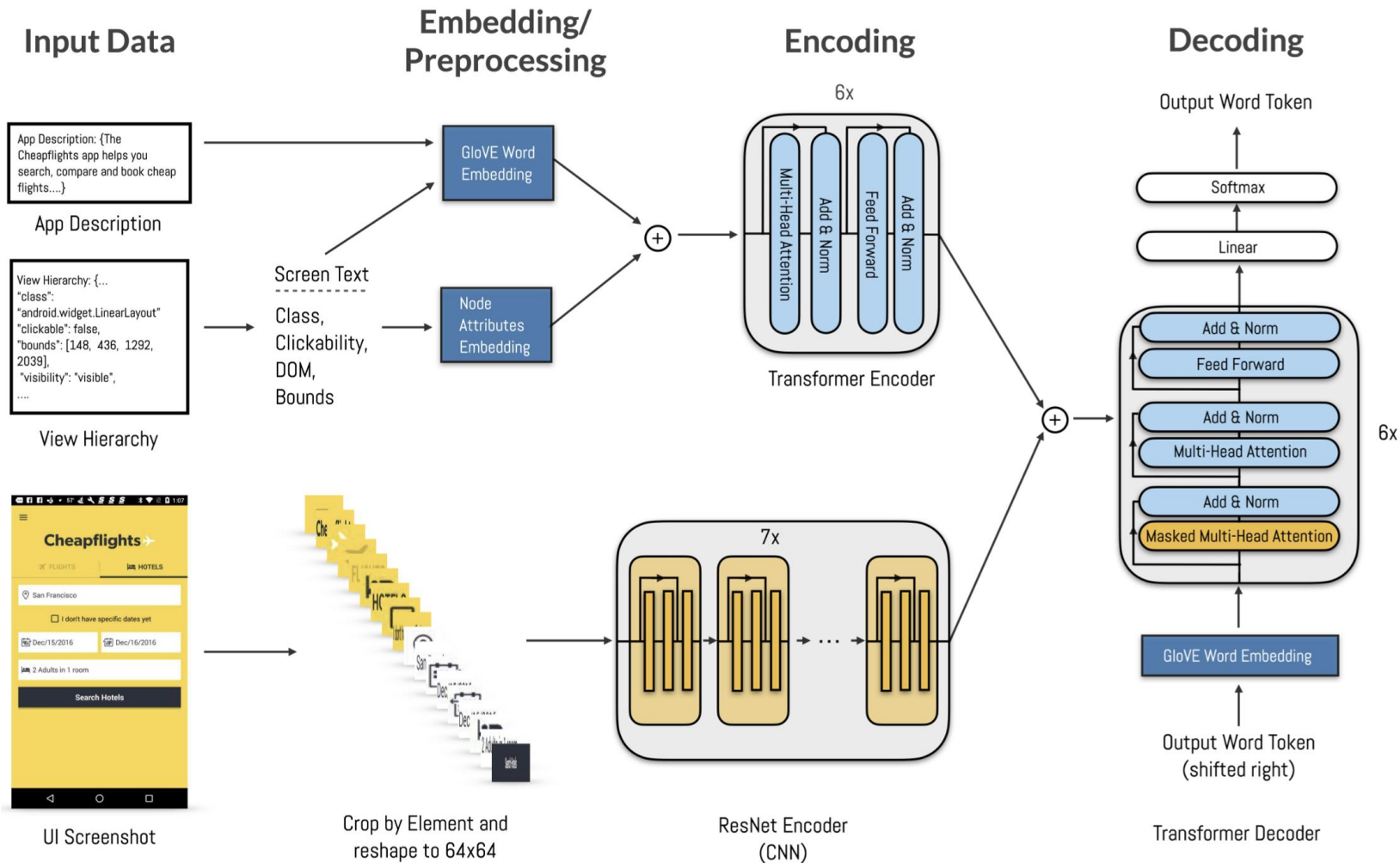
Screen2Words: Automatic Mobile UI Summarization

With Multimodal Master Learning



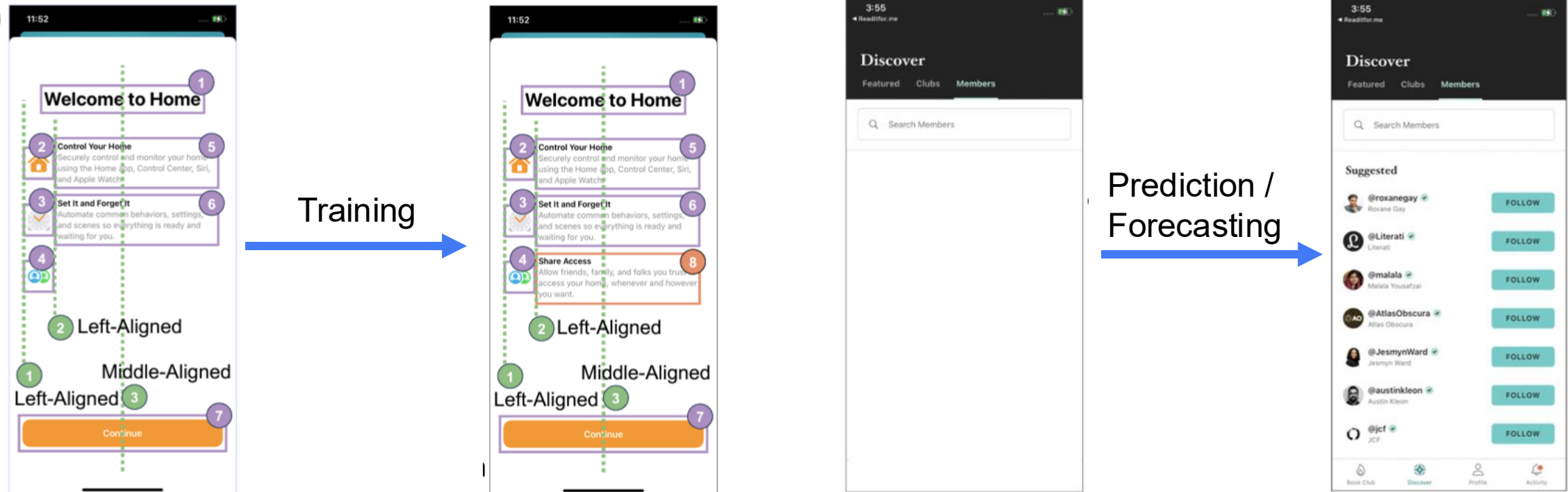
Bryan Wang, Gang Li, Xin Zhou, Zhourong Chen, Tovi Grossman, and Yang Li. 2021. Screen2Words: Automatic Mobile UI Summarization with Multimodal Learning. In The 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21). Association for Computing Machinery, New York, NY, USA, 498–510. <https://doi.org/10.1145/3472749.3474765>

Screen2Words: Automatic Mobile UI Summarization With MobileNet Mastering style



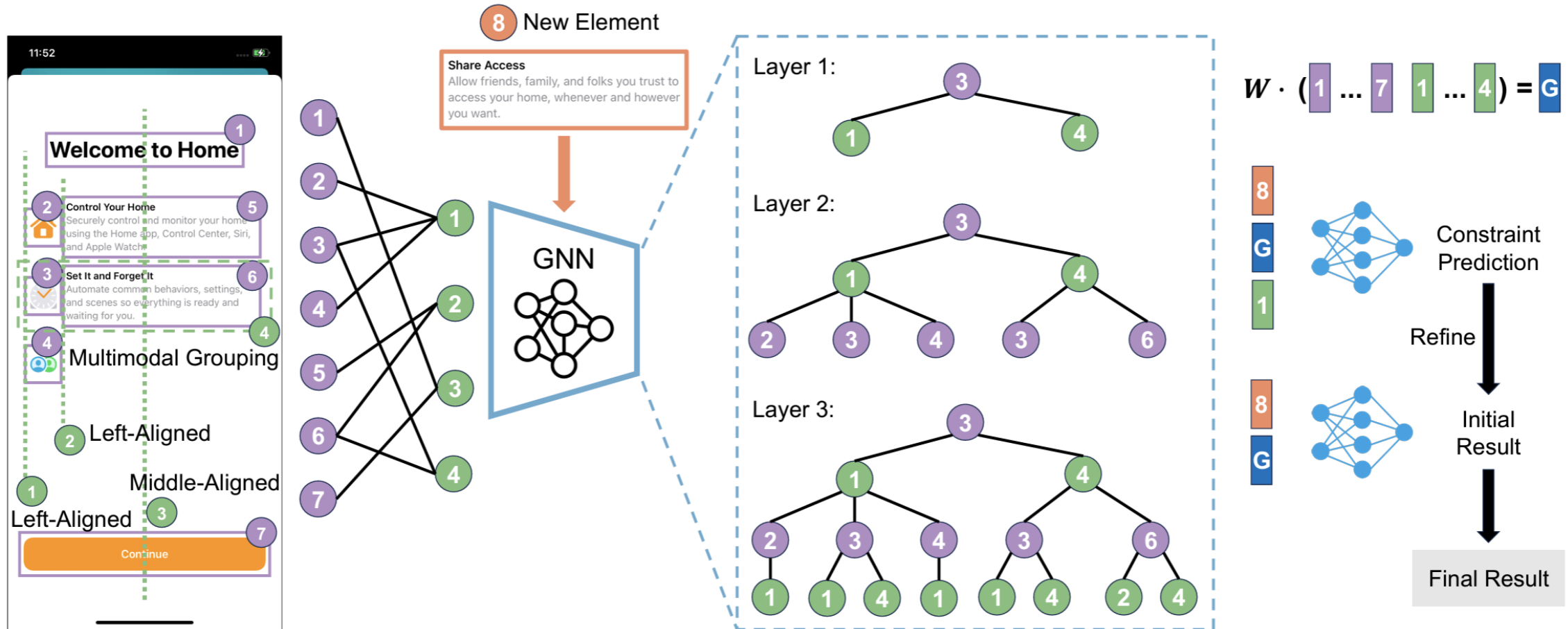
Graph4GUI: Graph Neural Networks for Representing Graphical User Interfaces

Graphicabits Master interface



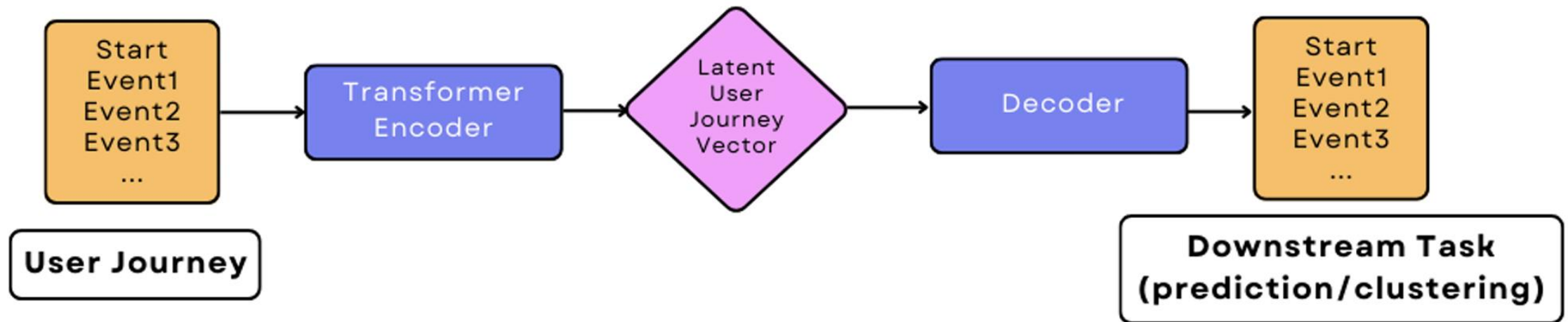
Yue Jiang, Changkong Zhou, Vikas Garg, and Antti Oulasvirta. 2024. Graph4GUI: Graph Neural Networks for Representing Graphical User Interfaces. In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 988, 1–18. <https://doi.org/10.1145/3613904.3642822>

Graph4GUI: Graph Neural Networks for Representing Graphical User Interfaces



Yue Jiang, Changkong Zhou, Vikas Garg, and Antti Oulasvirta. 2024. Graph4GUI: Graph Neural Networks for Representing Graphical User Interfaces. In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 988, 1–18. <https://doi.org/10.1145/3613904.3642822>

UserJourney2Vector: Enterprise Application of Transformer Models to User Telemetry Data



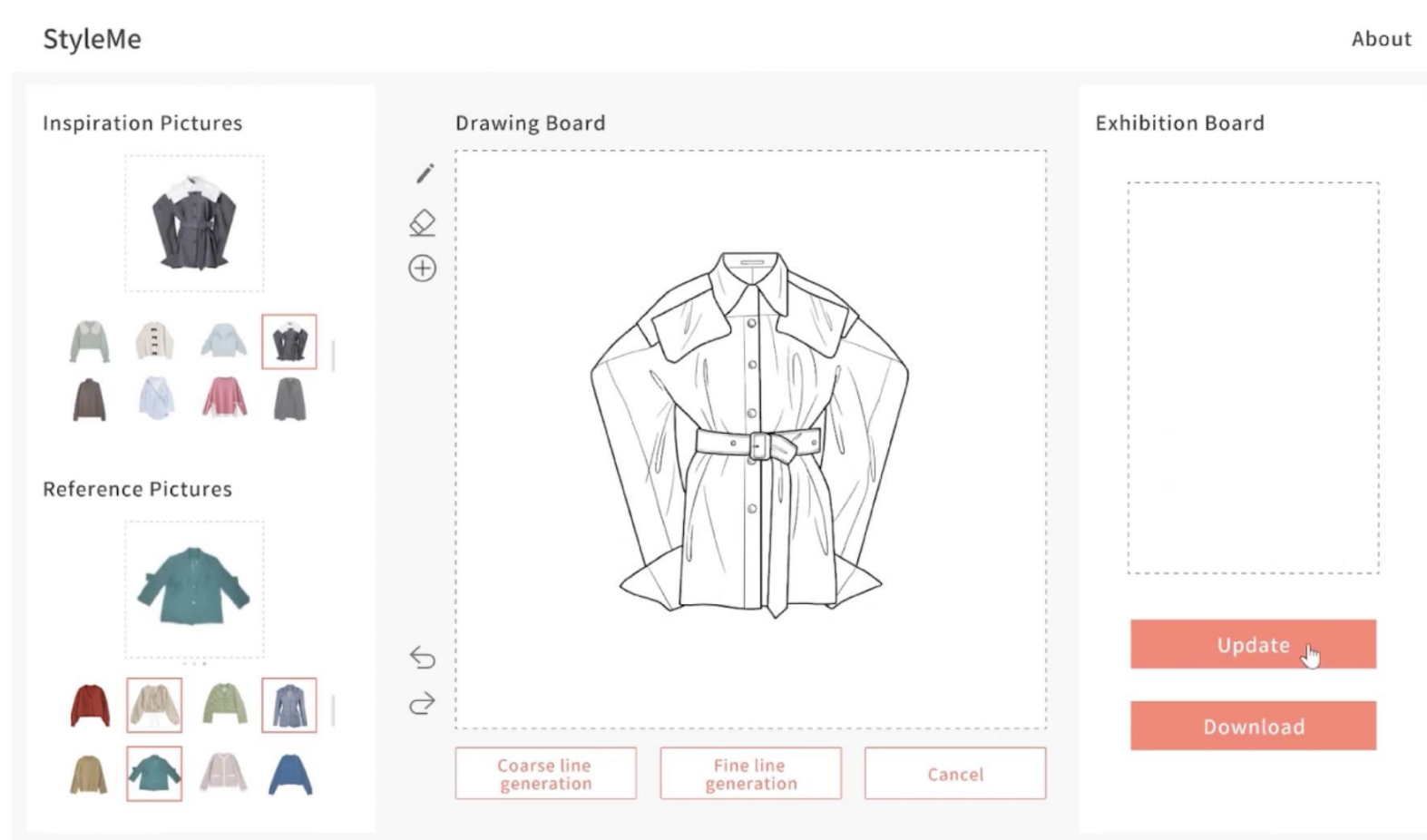
Sara Tandon and Dylan Contris. 2024. UserJourney2Vector: Enterprise Application of Transformer Models on User Telemetry Data. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 538, 1–7. <https://doi.org/10.1145/3613905.3637134>

StyleMe: Towards Intelligent Fashion Generation with DesignerStyleMaster title style



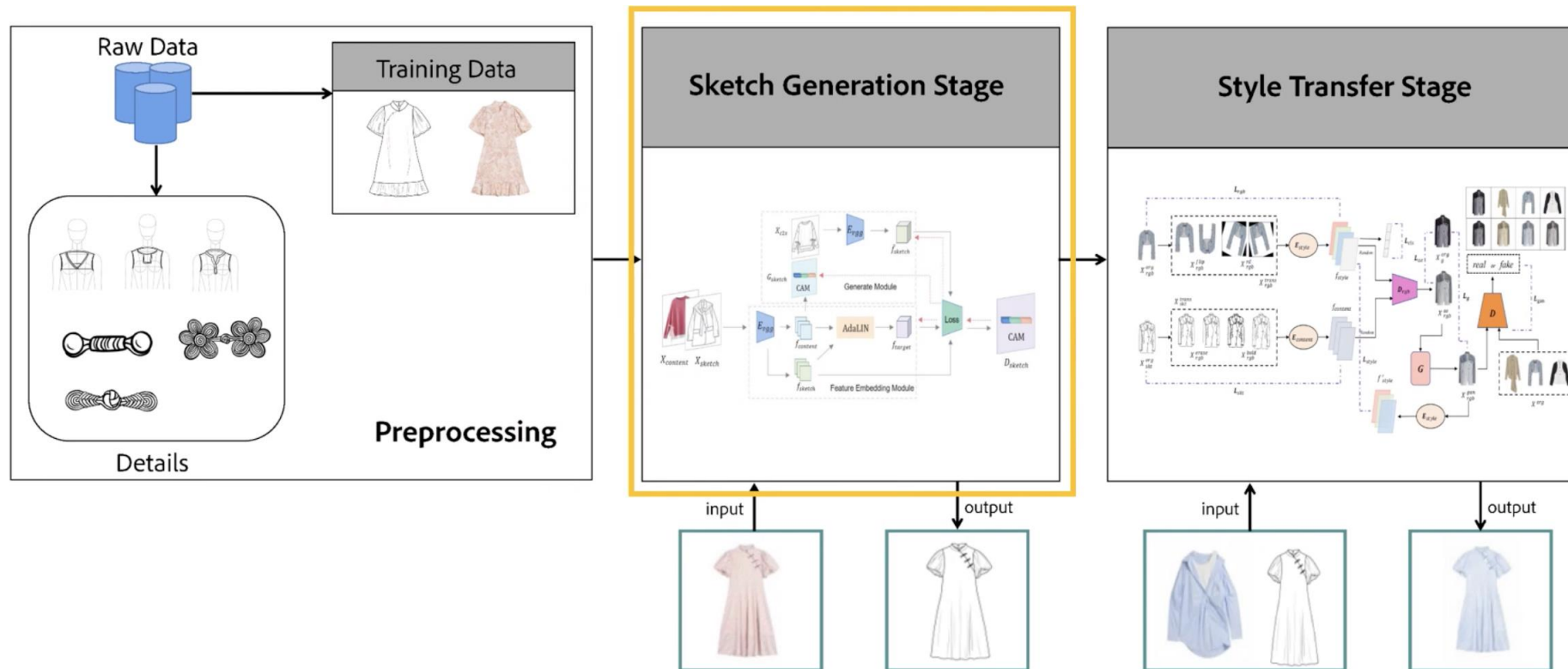
Sara Tandon and Dylan Contris. 2024. UserJourney2Vector: Enterprise Application of Transformer Models on User Telemetry Data. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 538, 1–7. <https://doi.org/10.1145/3613905.3637134>

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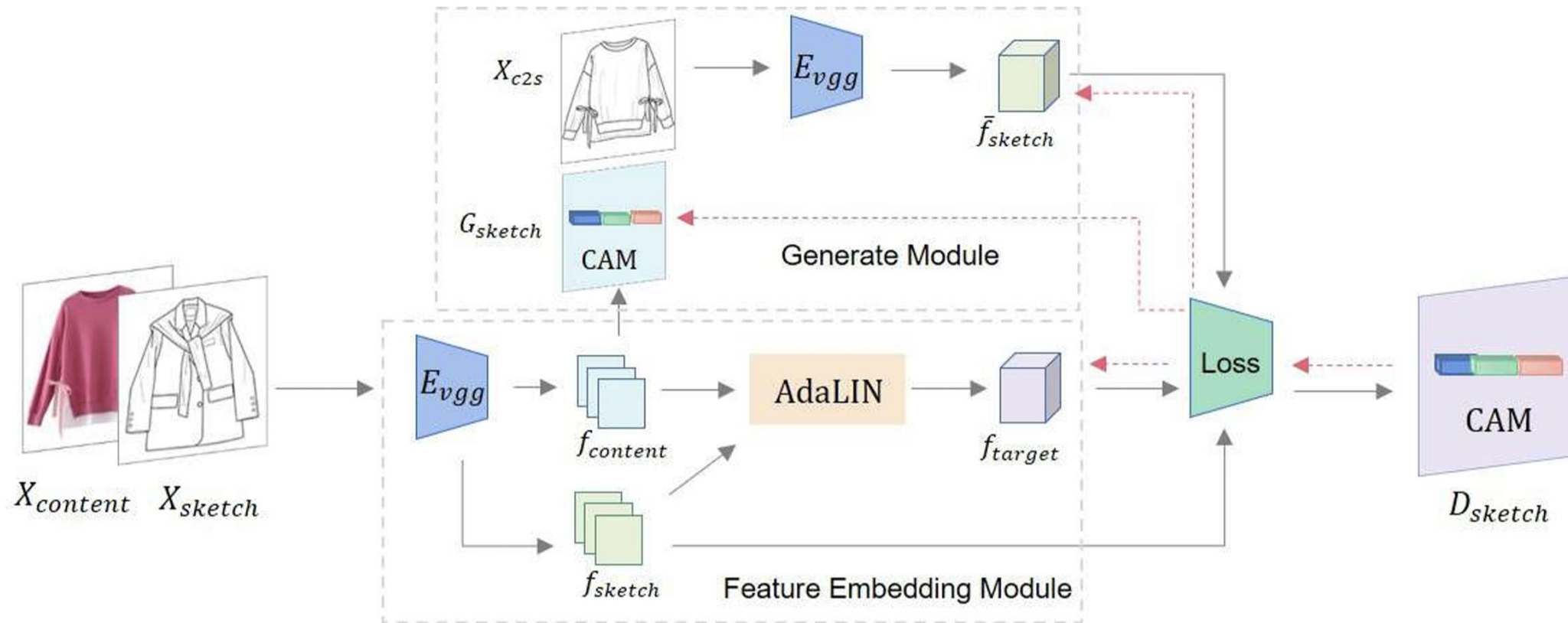
StyleMe: Towards Intelligent Fashion Generation with Designer Style Master title style



Sara Tandon and Dylan Contris. 2024. UserJourney2Vector: Enterprise Application of Transformer Models on User Telemetry Data. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 538, 1–7. <https://doi.org/10.1145/3613905.3637134>

StyleMe: Towards Intelligent Fashion Generation with Designer Style Master title style

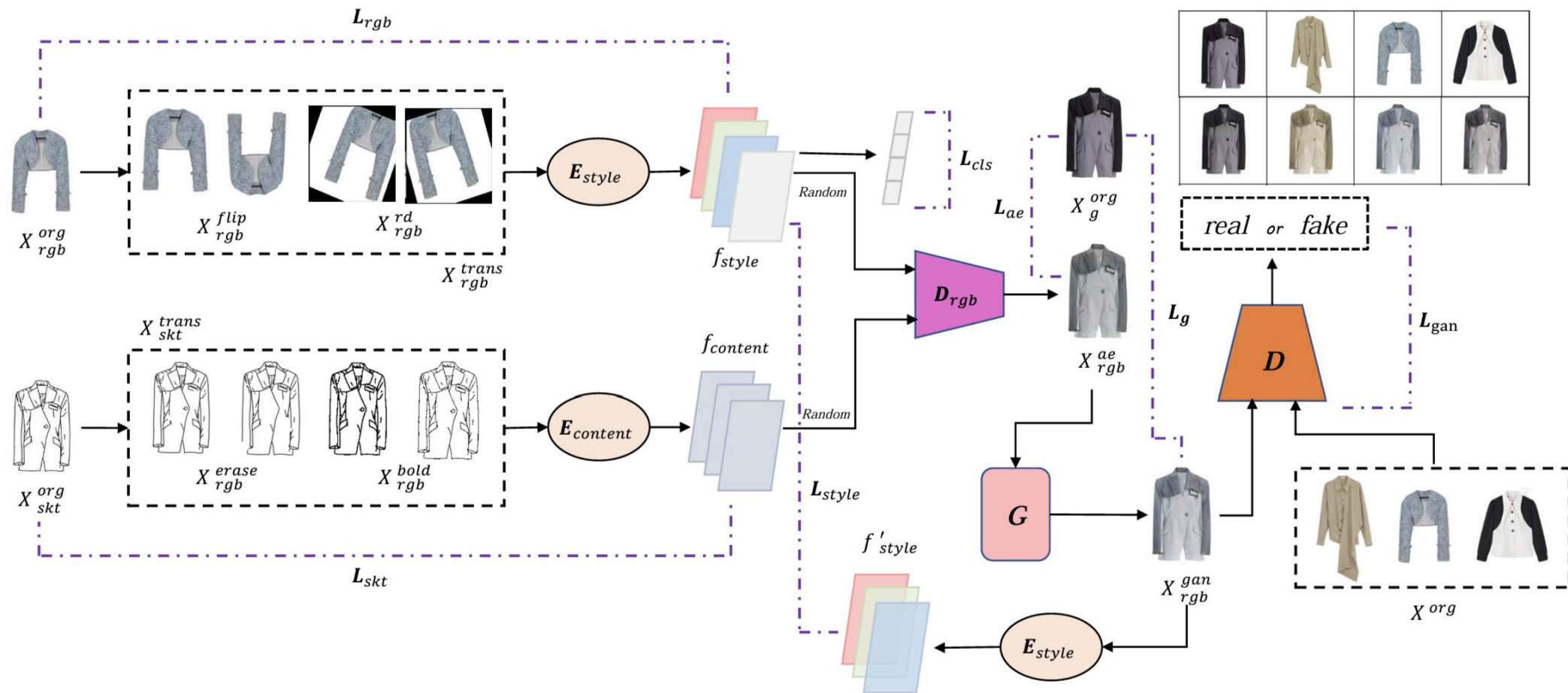
: An overview of personalized sketch automatic generation network architecture.



Sara Tandon and Dylan Contris. 2024. UserJourney2Vector: Enterprise Application of Transformer Models on User Telemetry Data. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 538, 1–7. <https://doi.org/10.1145/3613905.3637134>

StyleMe: Towards Intelligent Fashion Generation with Designer's Master title style

An overview of style transfer generation network architecture.



Itsy-Bits: Fabrication and Recognition of 3D-Printed Tangibles with Small Footprints on Capacitive Touchscreens



Figure 1: Itsy-Bits recognizes 3D-printed tangibles as small as a fingertip via the capacitive image of an embedded conductive shape (A). This opens up a variety of tangible user interfaces on the most common form factors of touchscreens, such as individualized interactive board games (B) or a more tangible learning experience (C).

Martin Schmitz, Florian Müller, Max Mühlhäuser, Jan Riemann, and Huy Viet Viet Le. 2021. Itsy-Bits: Fabrication and Recognition of 3D-Printed Tangibles with Small Footprints on Capacitive Touchscreens. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, Article 419, 1–12. DOI: <https://doi.org/10.1145/3411764.3445502>

Itsy-Bits: Fabrication and Recognition of 3D-Printed Tangibles with Small Footprints on Capacitive Touchscreens

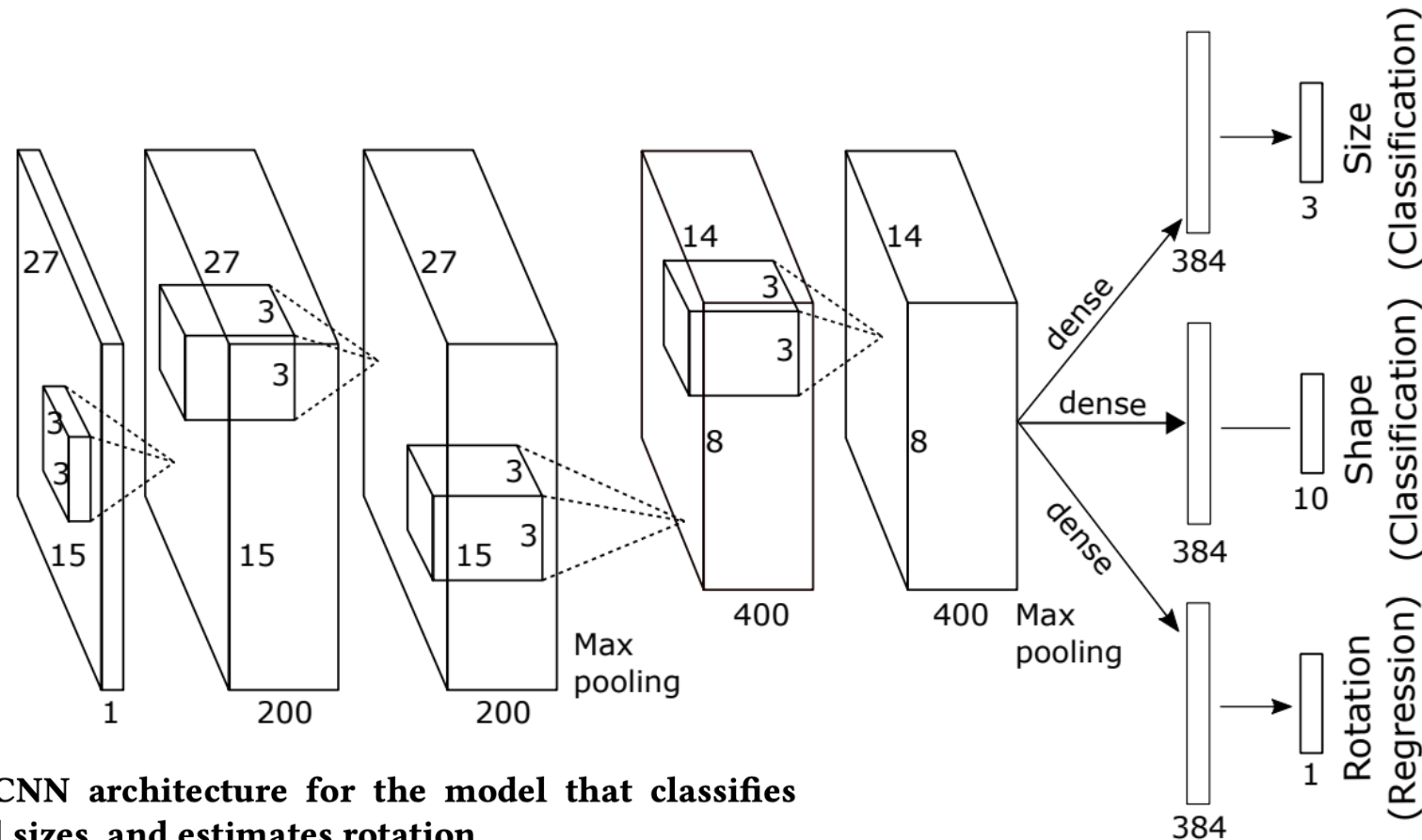
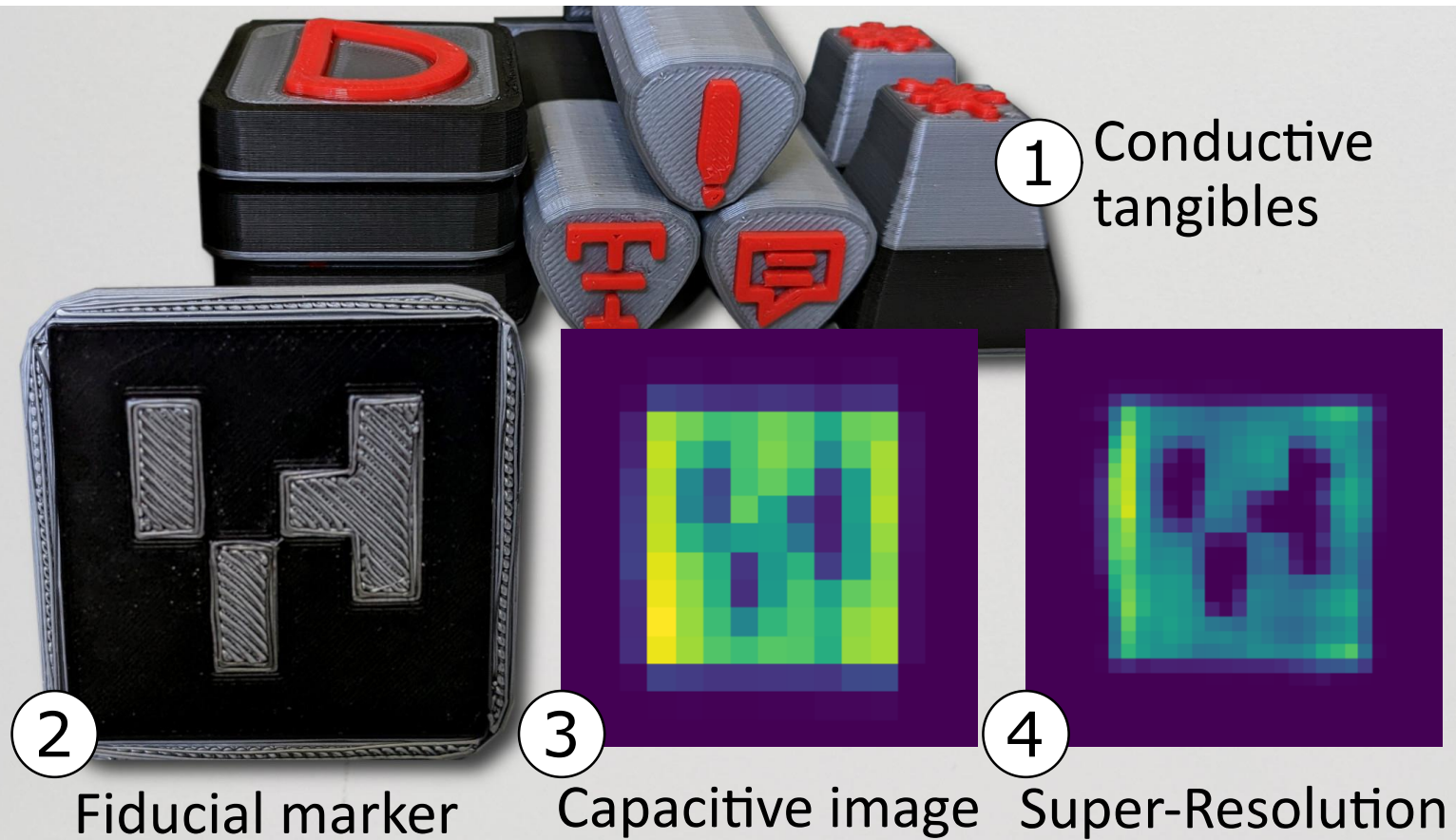


Figure 7: CNN architecture for the model that classifies shapes and sizes, and estimates rotation.

Martin Schmitz, Florian Müller, Max Mühlhäuser, Jan Riemann, and Huy Viet Viet Le. 2021. Itsy-Bits: Fabrication and Recognition of 3D-Printed Tangibles with Small Footprints on Capacitive Touchscreens. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, Article 419, 1–12. DOI: <https://doi.org/10.1145/3411764.3445502>

Super Resolution + GAN



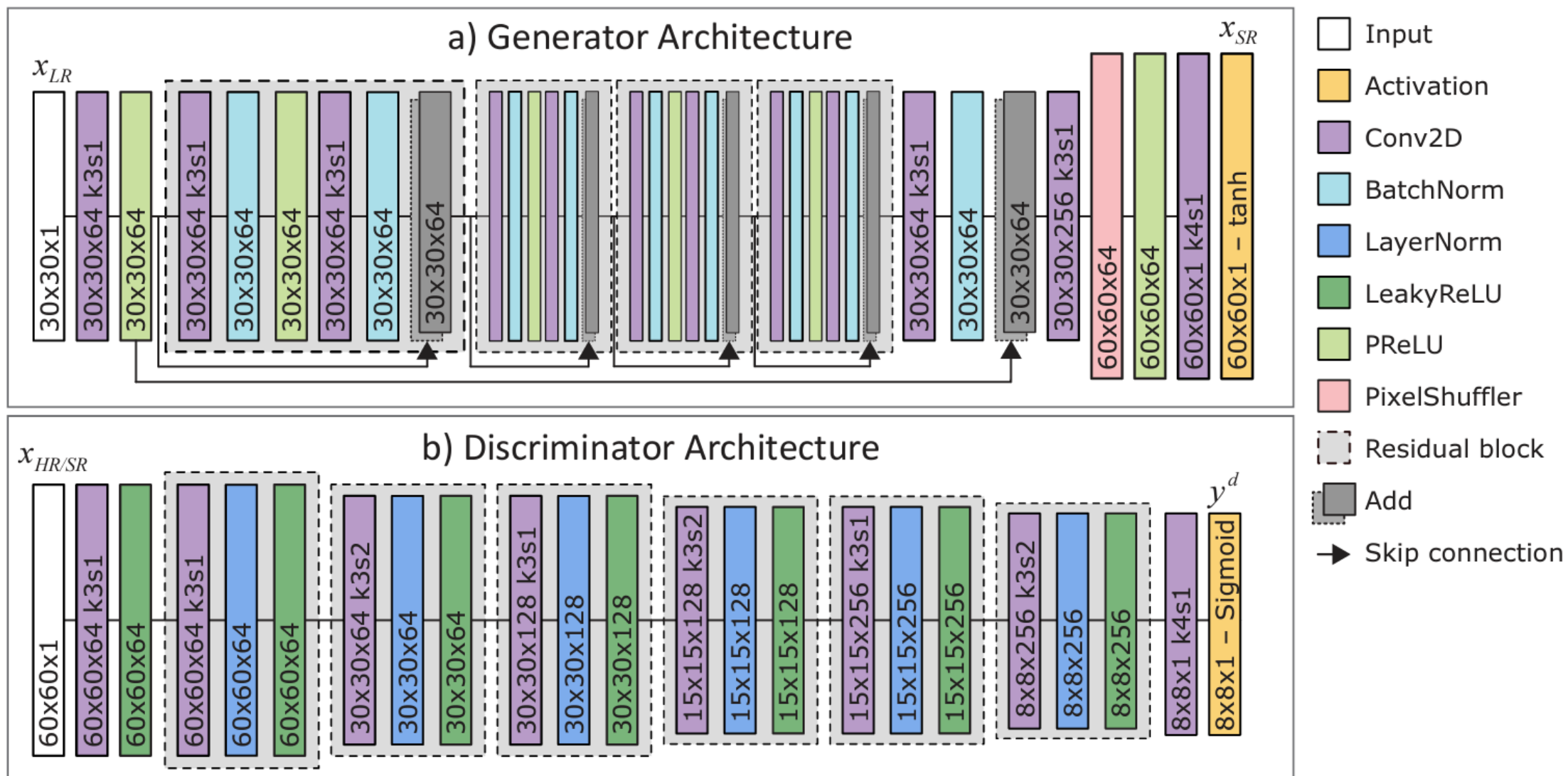


Figure 5: a) The architecture of the Generator with 1,002,433 parameters. b) The architecture of the Discriminator with 1,187,073 parameters.

CapContact: Super-resolution Contact Areas from Capacitive Touchscreens

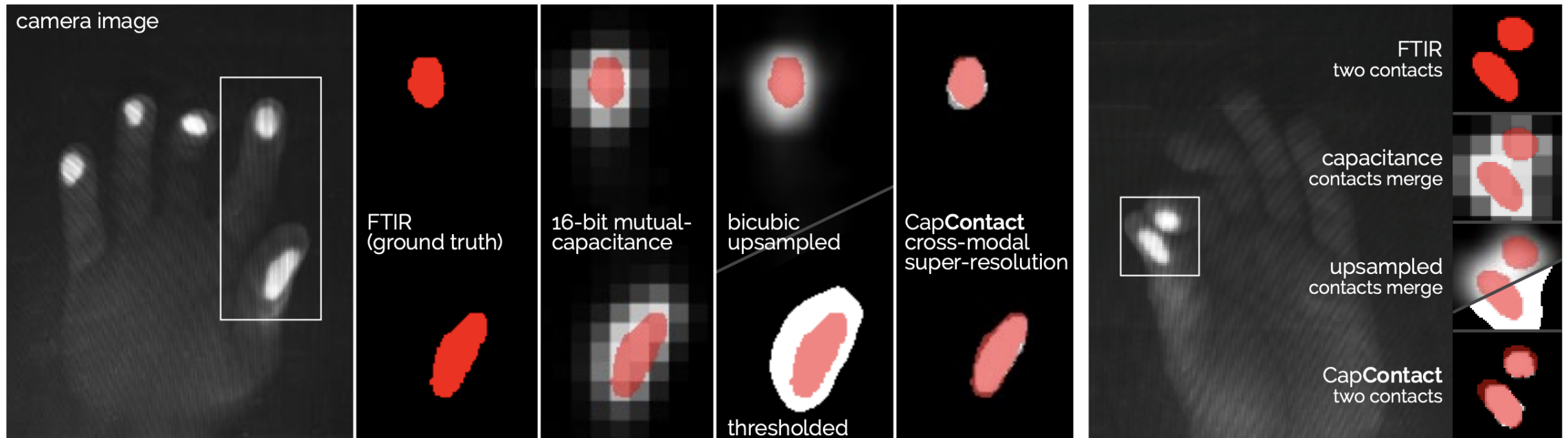


Figure 1: CapContact reconstructs a *high-resolution contact area* between the user’s fingers and the touch surface from the typical low-resolution capacitive sensor. We designed and trained an $8\times$ cross-modality super-resolution convolutional neural network on pairs of regular-resolution mutual-capacitance sensor images and ground-truth contact maps from a frustrated total internal reflection setup (FTIR, contact areas overlaid in red in each image). CapContact produces significantly more accurate contact areas than the baseline of thresholded bicubic upsampling on the capacitive images. Unlike existing methods, CapContact can discriminate closely adjacent touches into *separate* contact areas that merge in raw capacitive recordings.

Paul Streli and Christian Holz. 2021. CapContact: Super-resolution Contact Areas from Capacitive Touchscreens. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 289, 1–14. DOI: <https://doi.org/10.1145/3411764.3445621>

CapContact: Super-resolution Contact Areas from Capacitive Touchscreens

Generator

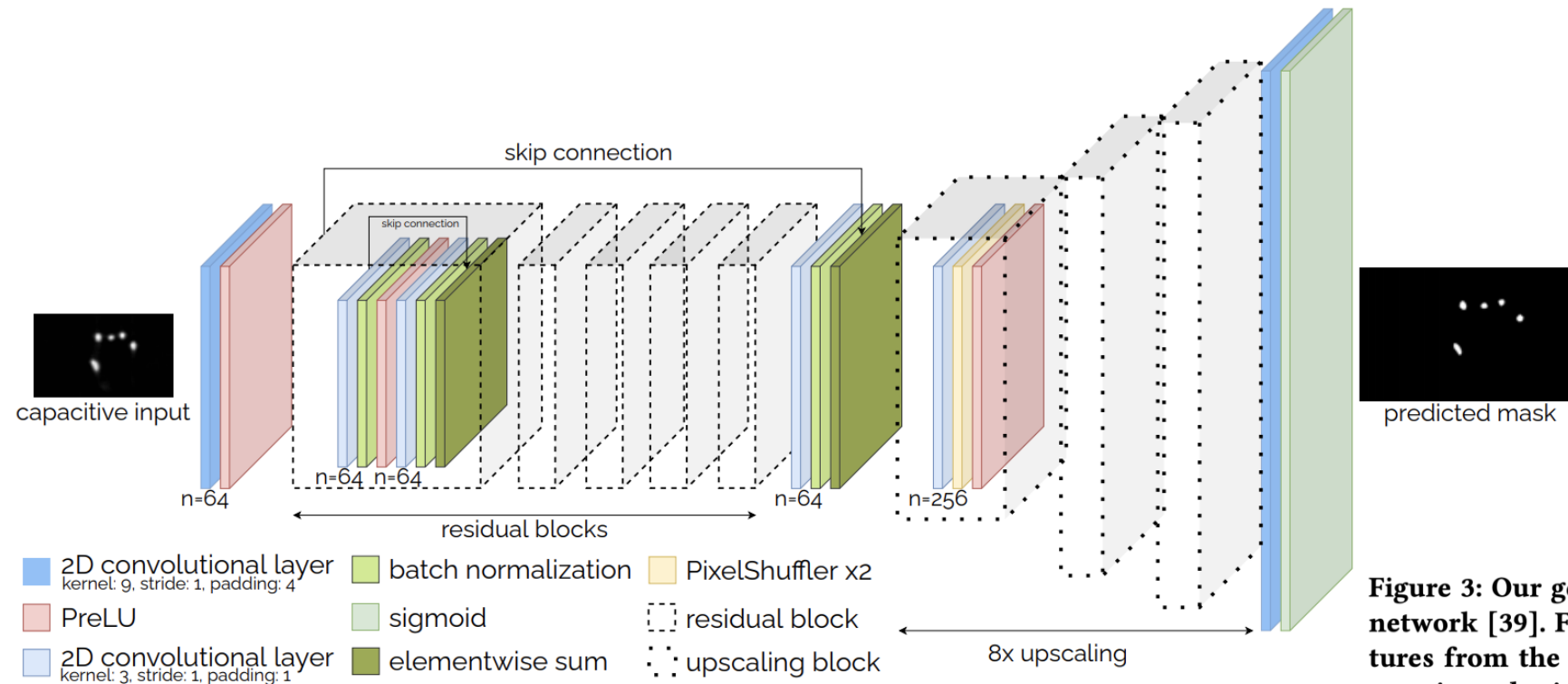


Figure 3: Our generator is a VGG-style fully-convolutional network [39]. Five residual blocks extract the relevant features from the low-resolution capacitive input. Three consecutive sub-pixel convolution layers then upsample it 8×.

TapNet: The Design, Training, Implementation, and Applications of a Multi-Task Learning CNN for Off-Screen Mobile Input

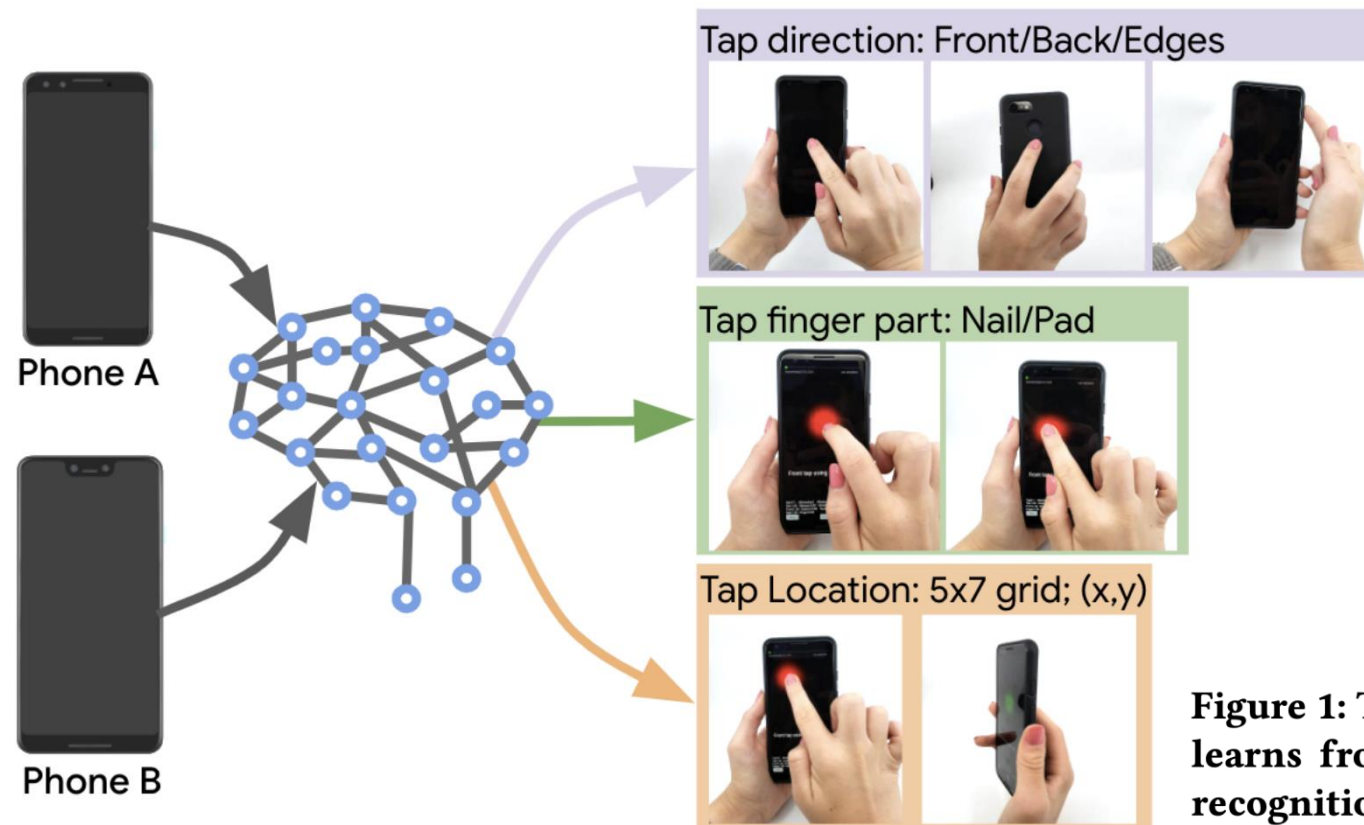


Figure 1: TapNet is a one-model-for-all solution that jointly learns from cross-device data and performs multiple tap recognition tasks at a time.

Michael Xuelin Huang, Yang Li, Nazneen Nazneen, Alexander Chao, and Shumin Zhai. 2021. TapNet: The Design, Training, Implementation, and Applications of a Multi-Task Learning CNN for Off-Screen Mobile Input. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 282, 1–11. DOI: <https://doi.org/10.1145/3411764.3445626>

TapNet: The Design, Training, Implementation, and Applications of a Multi-Task Learning CNN for Off-Screen Mobile Input

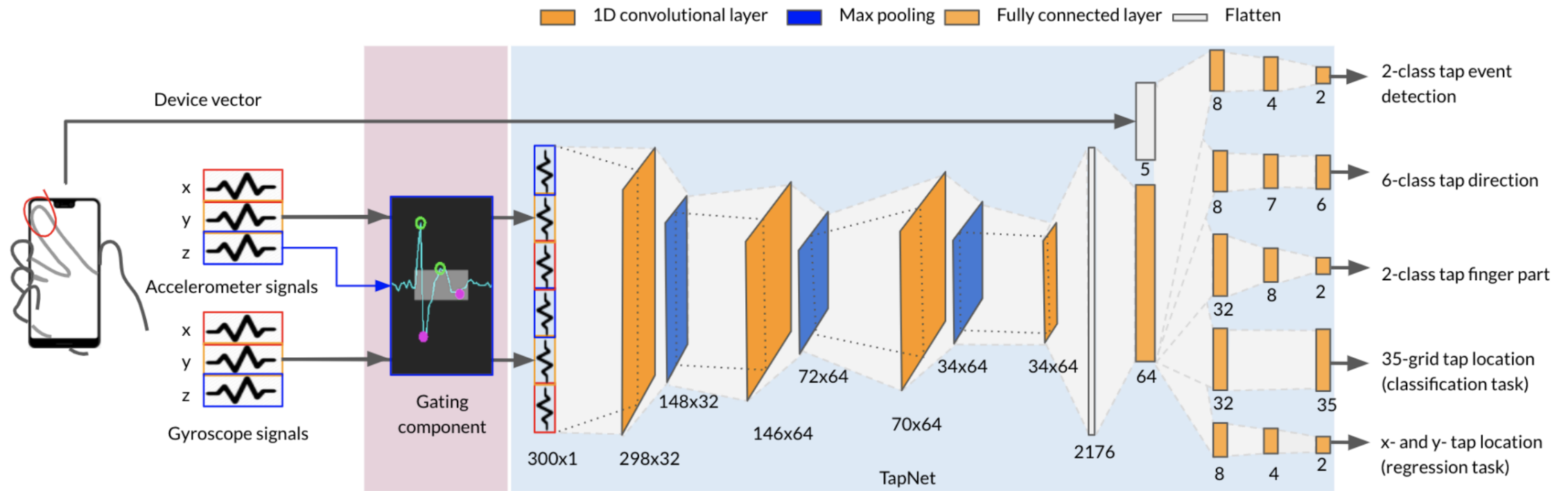


Figure 2: Accelerometer and gyroscope signals are processed in the Gating component and passed to TapNet if met set criteria. Using phone form factor (embedded in the device vector) as auxiliary information, TapNet jointly recognizes multiple tap properties, including tap event, direction, finger part, and location.

Michael Xuelin Huang, Yang Li, Nazneen Nazneen, Alexander Chao, and Shumin Zhai. 2021. TapNet: The Design, Training, Implementation, and Applications of a Multi-Task Learning CNN for Off-Screen Mobile Input. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 282, 1–11. DOI: <https://doi.org/10.1145/3411764.3445626>

Assessing Task Difficulty through Physiological Representation of Spontaneous Blinking

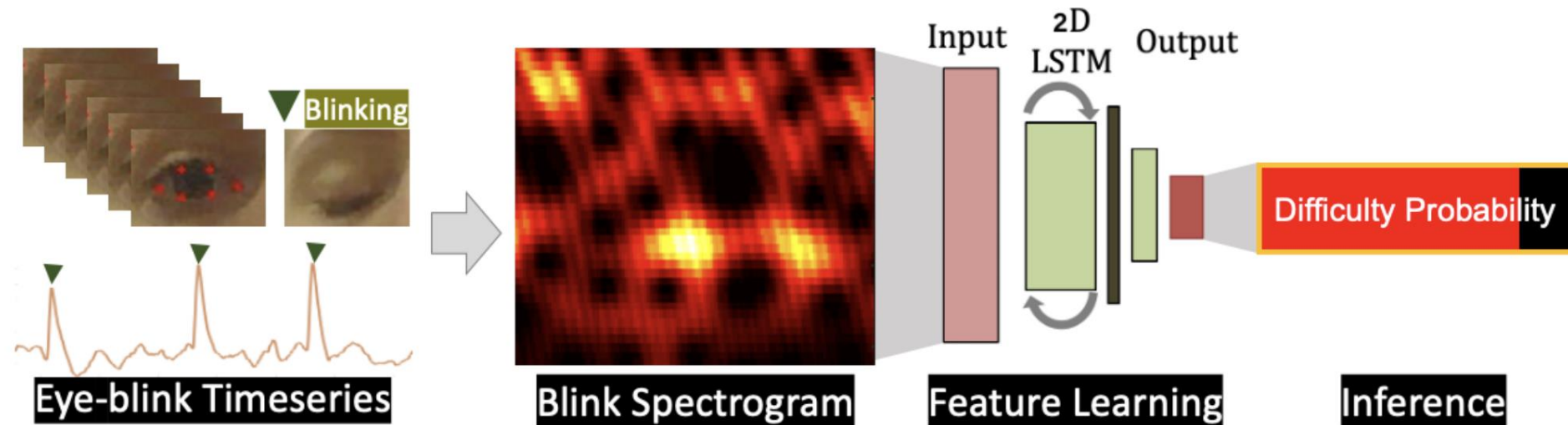


Figure 1: *Rethinking Eye-Blink* is a novel strategy to monitor task difficulty and mental workload continuously and automatically through a physiological representation of a user's spontaneous blinking. This approach uses a standard RGB camera to extract eye-blink timeseries which we represent on time-frequency spectrograms for automatic feature learning.

Youngjun Cho. 2021. Rethinking Eye-blink: Assessing Task Difficulty through Physiological Representation of Spontaneous Blinking. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 721, 1–12. DOI: <https://doi.org/10.1145/3411764.3445577>

DeepTake: Prediction of Driver Takeover Behavior using Multimodal Data

Feature Engineering & 3-hidden layer

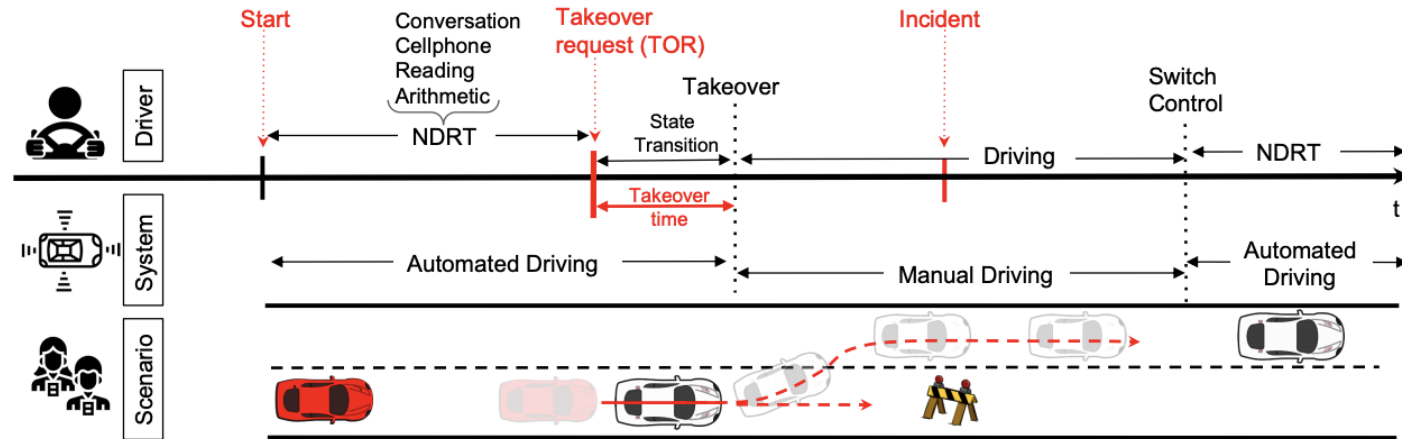


Figure 3: A schematic view of an example of a takeover situation used in our study, consisting of: 1) takeover timeline associated with participants' course of action; 2) system status; and 3) takeover situation. The vehicle was driven in the automated mode to the point after the TOR initiation and transitioning preparation period. The ego vehicle is shown in red and the lead car is white. When the Ego vehicle reaches its limits, the system may initiate (true alarm) or fail (no alarm) to initiate the TOR, and the driver takes the control back from the automated system.

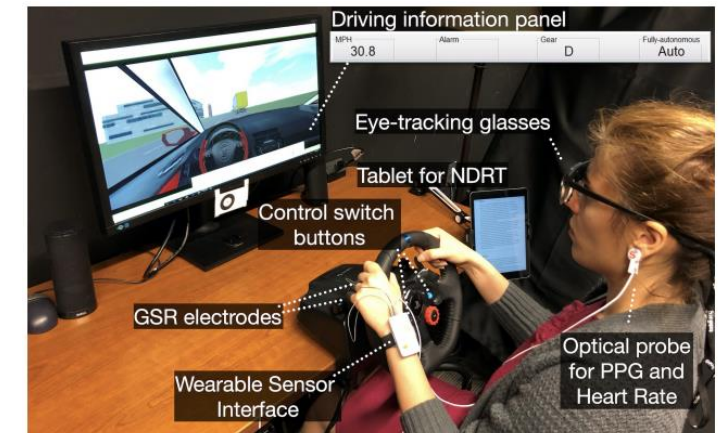
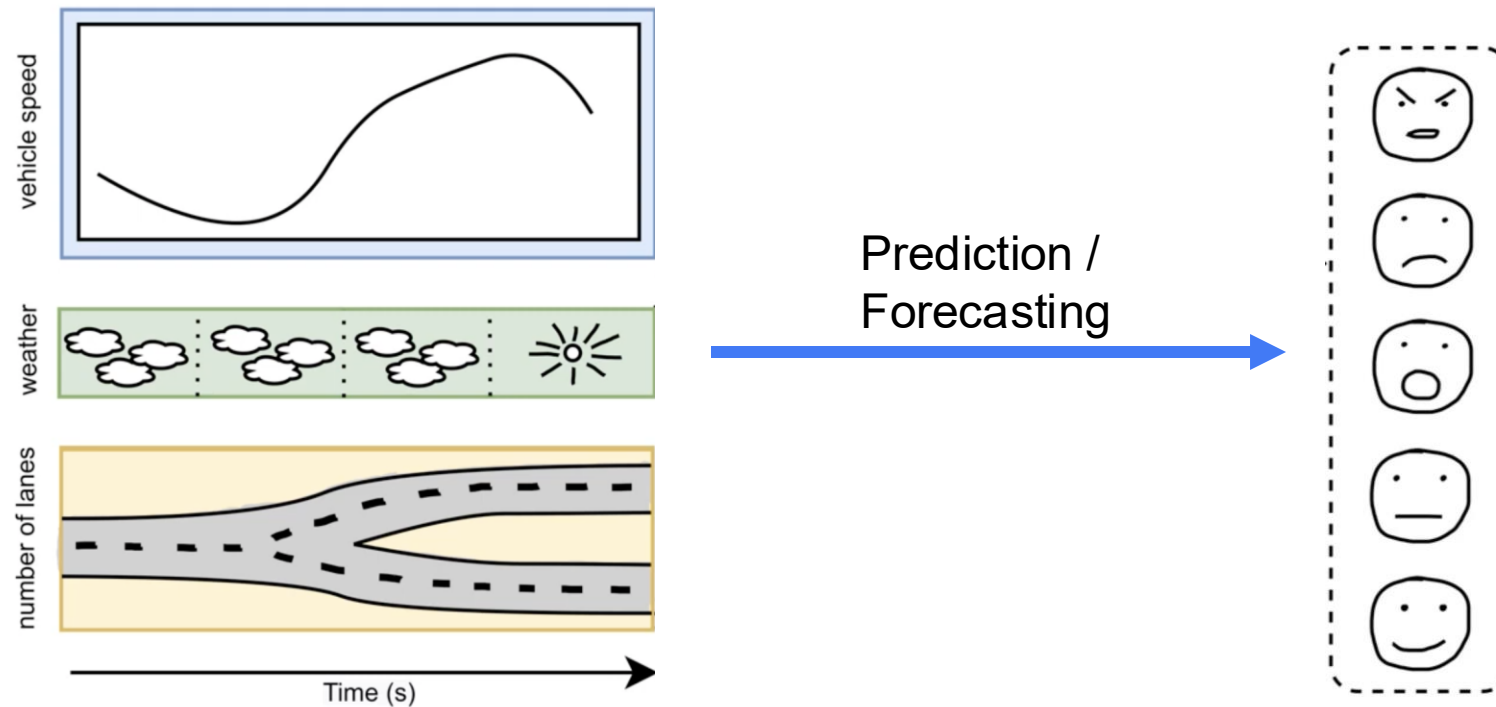


Figure 2: User study setup. This custom driving simulator consists of a 30-inch monitor, a Logitech G29 steering wheel, and 10.5-inch Apple iPad Air on which the non-driving tasks are displayed. For switching between the automated and manual control of the vehicle, the participant needs to press the two blue buttons on the steering wheel simultaneously. The participant wears a pair of eye-tracking glasses, and a wearable device with GSR and PPG sensors for the biometrics acquisition.

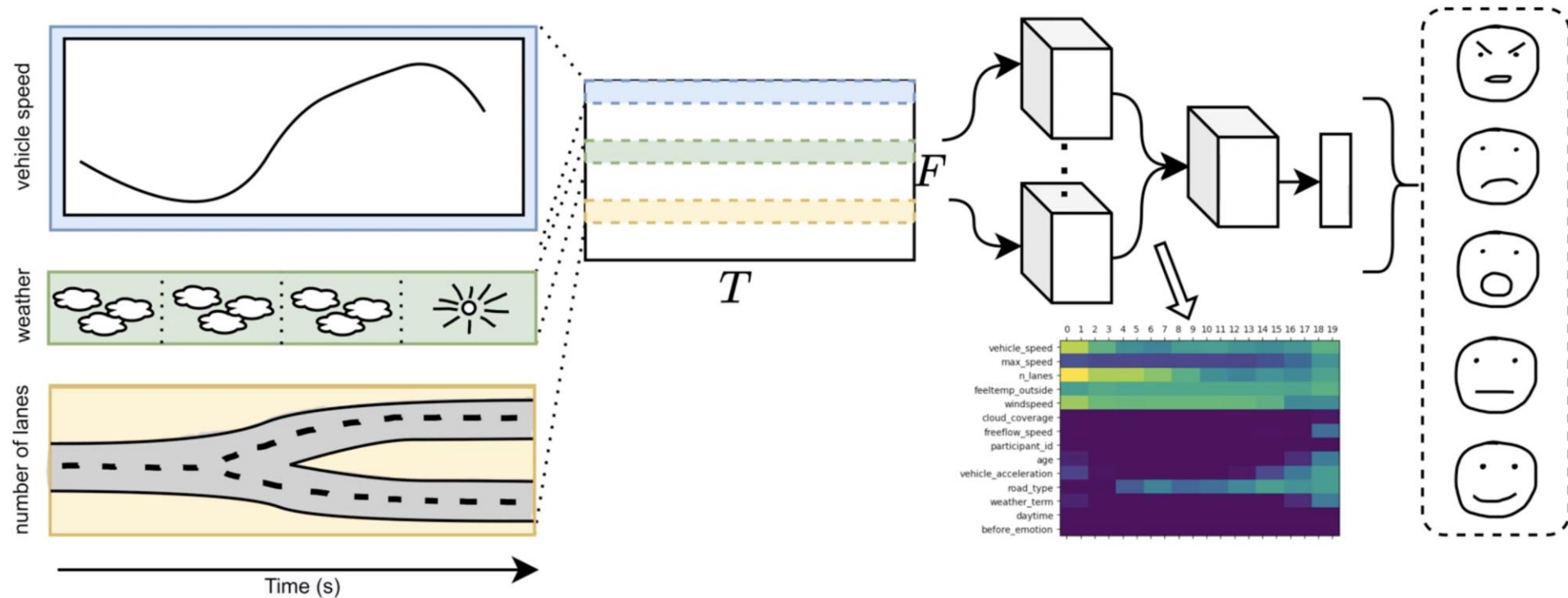
Erfan Pakdamanian, Shili Sheng, Sonia Baee, Seongkook Heo, Sarit Kraus, and Lu Feng. 2021. DeepTake: Prediction of Driver Takeover Behavior using Multimodal Data. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 103, 1–14. DOI: <https://doi.org/10.1145/3411764.3445563>

Interpretable Time-Dependent Convolutional Emotion Recognition with Contextual Data Streams



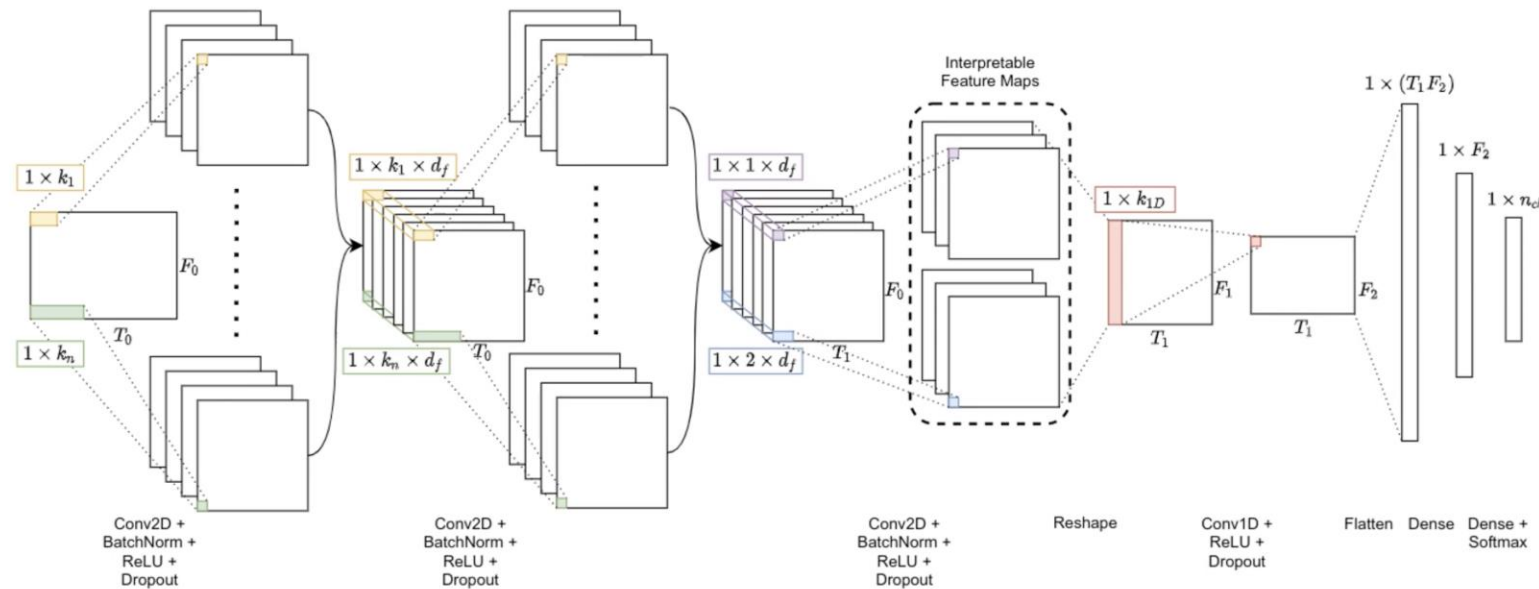
David Bethge, Constantin Patsch, Philipp Hallgarten, and Thomas Kosch. 2023. Interpretable Time-Dependent Convolutional Emotion Recognition with Contextual Data Streams. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 186, 1–9. <https://doi.org/10.1145/3544549.3585672>

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David Bethge, Constantin Patsch, Philipp Hallgarten, and Thomas Kosch. 2023. Interpretable Time-Dependent Convolutional Emotion Recognition with Contextual Data Streams. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 186, 1–9. <https://doi.org/10.1145/3544549.3585672>

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RadarNet: Efficient Gesture Recognition Technique Utilizing a Miniature Radar Sensor



Eiji Hayashi, Jaime Lien, Nicholas Gillian, Leonardo Giusti, Dave Weber, Jin Yamanaka, Lauren Bedal, and Ivan Poupyrev. 2021. RadarNet: Efficient Gesture Recognition Technique Utilizing a Miniature Radar Sensor. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 5, 1–14. DOI: <https://doi.org/10.1145/3411764.3445367>

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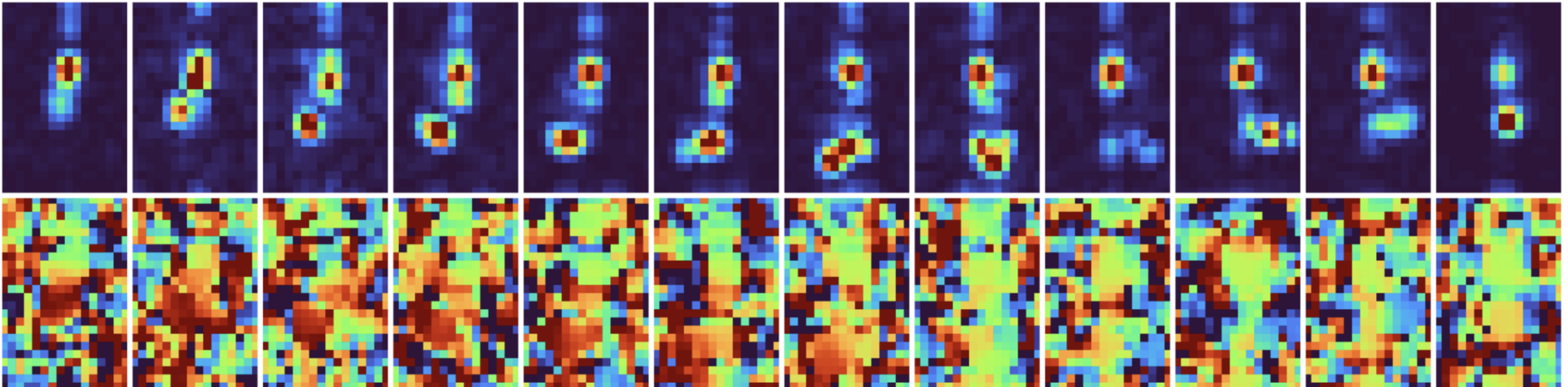


Figure 1: A time series visualization of a left swipe motion with absolute range Doppler maps (top) and interferometry maps (bottom). A hand moving close to a Soli chip round 6th frame, then, moving away. In the interferometry maps, the color of cells s at the same position as hand above changed from red to green, and to blue, showing the hand moving right to left.

Eiji Hayashi, Jaime Lien, Nicholas Gillian, Leonardo Giusti, Dave Weber, Jin Yamanaka, Lauren Bedal, and Ivan Poupyrev. 2021. RadarNet: Efficient Gesture Recognition Technique Utilizing a Miniature Radar Sensor. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 5, 1–14. DOI: <https://doi.org/10.1145/3411764.3445367>

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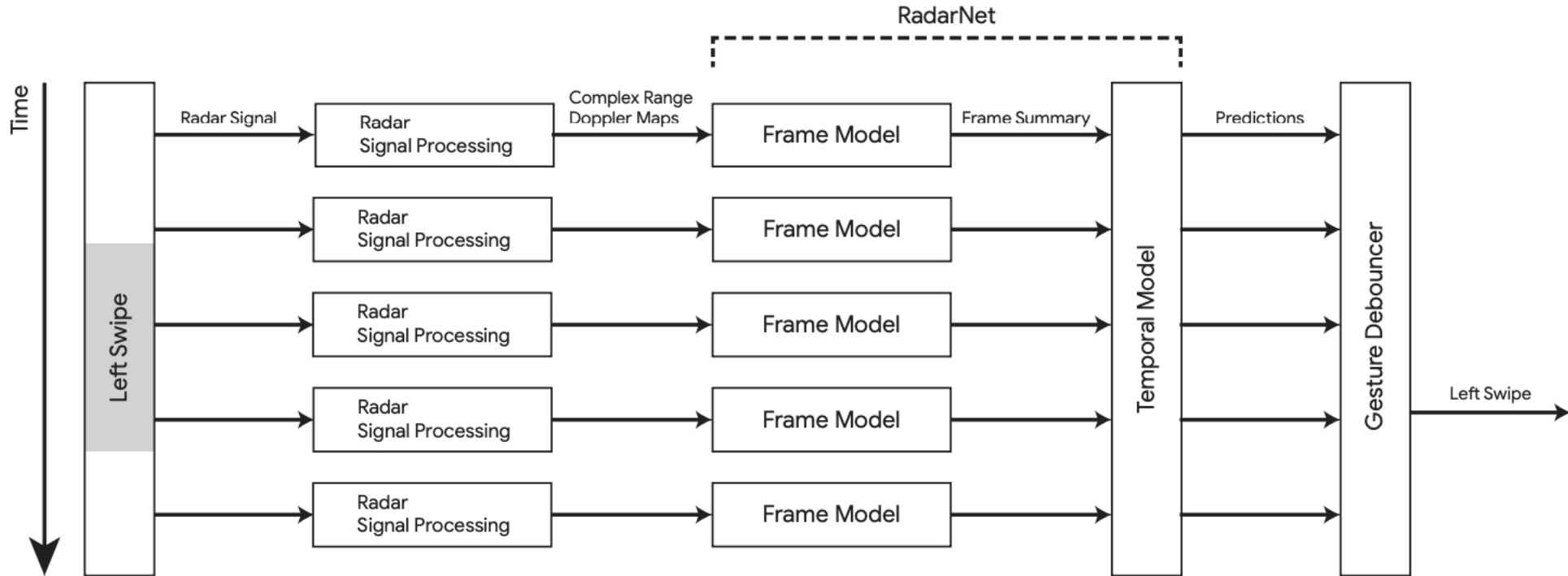
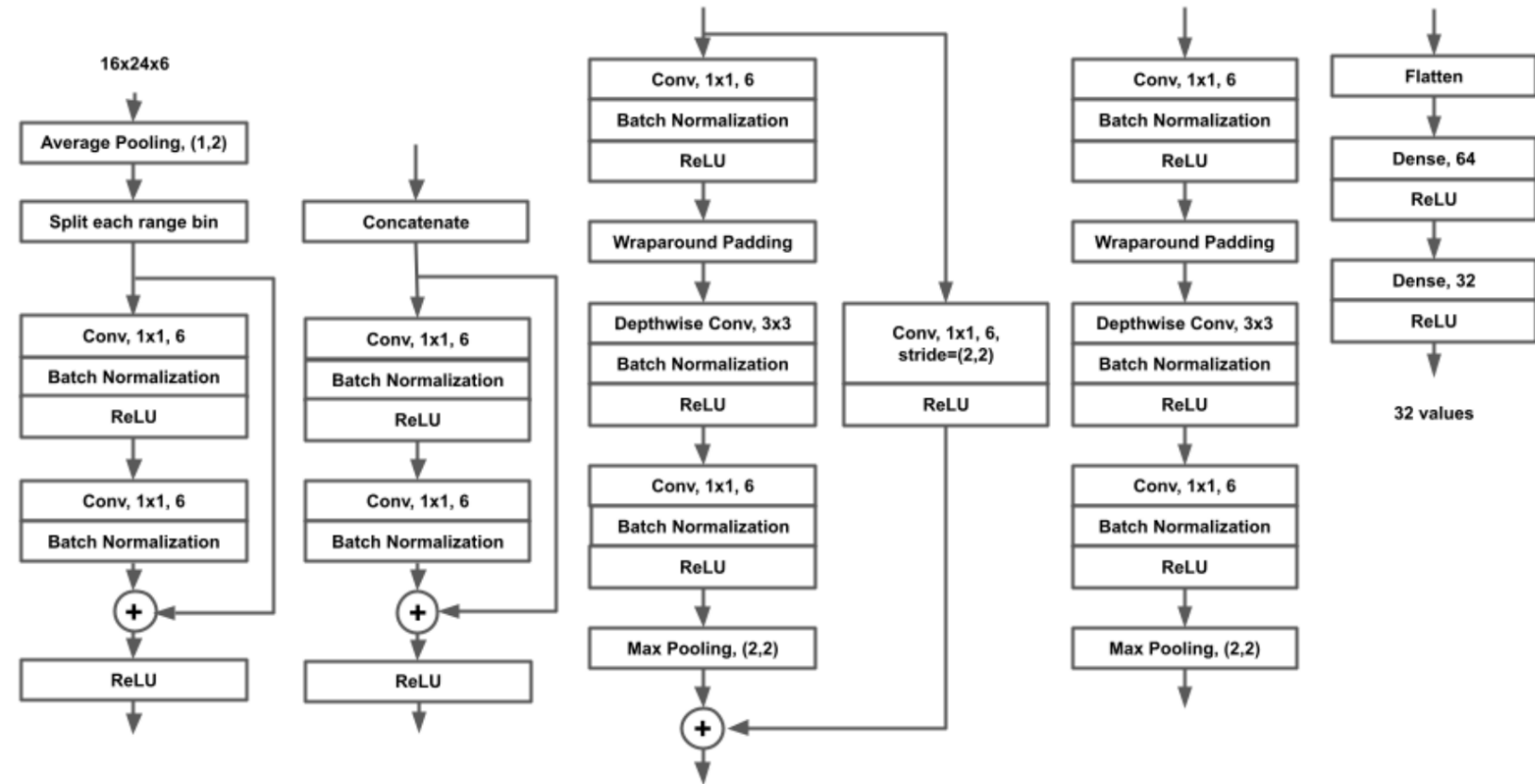


Figure 6: The pipeline applies signal processing algorithms to the radar signals to compute complex range Doppler maps for each burst. The complex range Doppler maps are converted into a summary consisting of 32 values with the frame model of the RadarNet. Then the summaries in the last 12 frames are processed by the temporal model of the RadarNet to make portrait, landscape, and omni predictions. Finally, the gesture debouncer output recognizes gestures.

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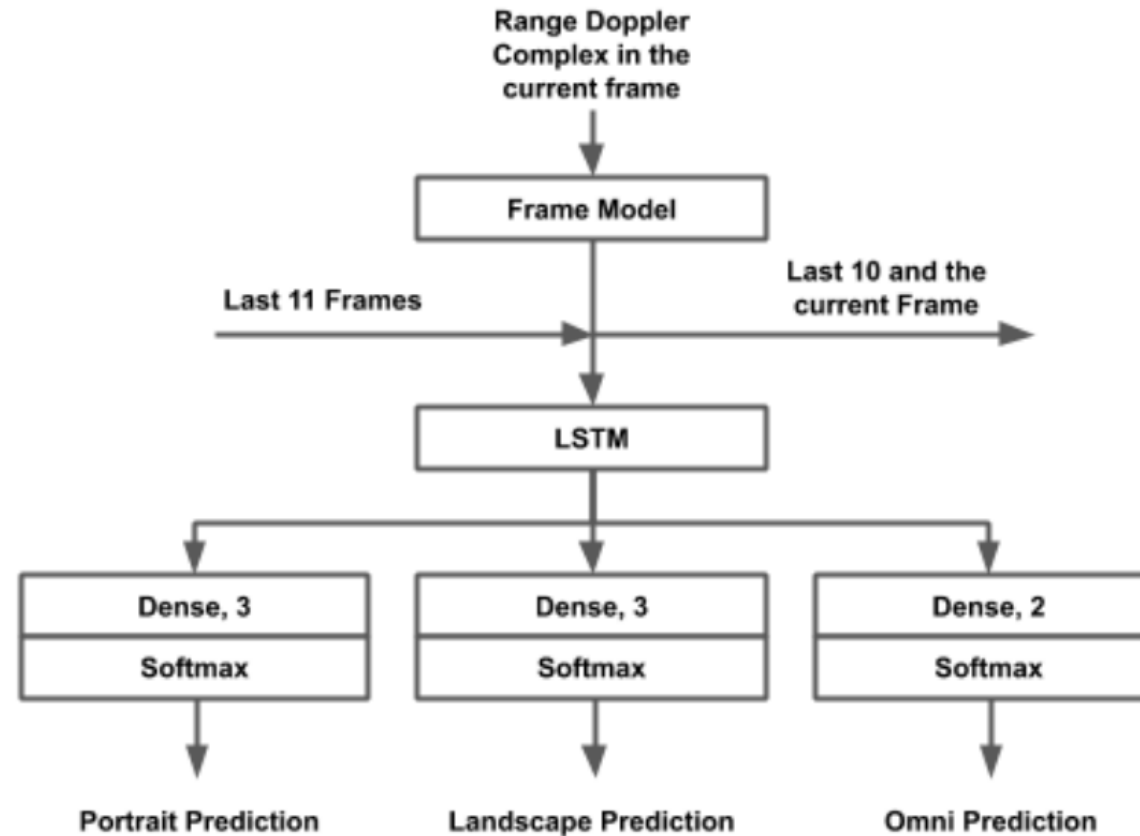
Fame Model



Eiji Hayashi, Jaime Lien, Nicholas Gillian, Leonardo Giusti, Dave Weber, Jin Yamanaka, Lauren Bedal, and Ivan Poupyrev. 2021. RadarNet: Efficient Gesture Recognition Technique Utilizing a Miniature Radar Sensor. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 5, 1–14. DOI: <https://doi.org/10.1145/3411764.3445367>

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Temporal Model



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Improving Viewing Experiences of First-Person Shooter Gameplays with Automatically-Generated Motion Effects

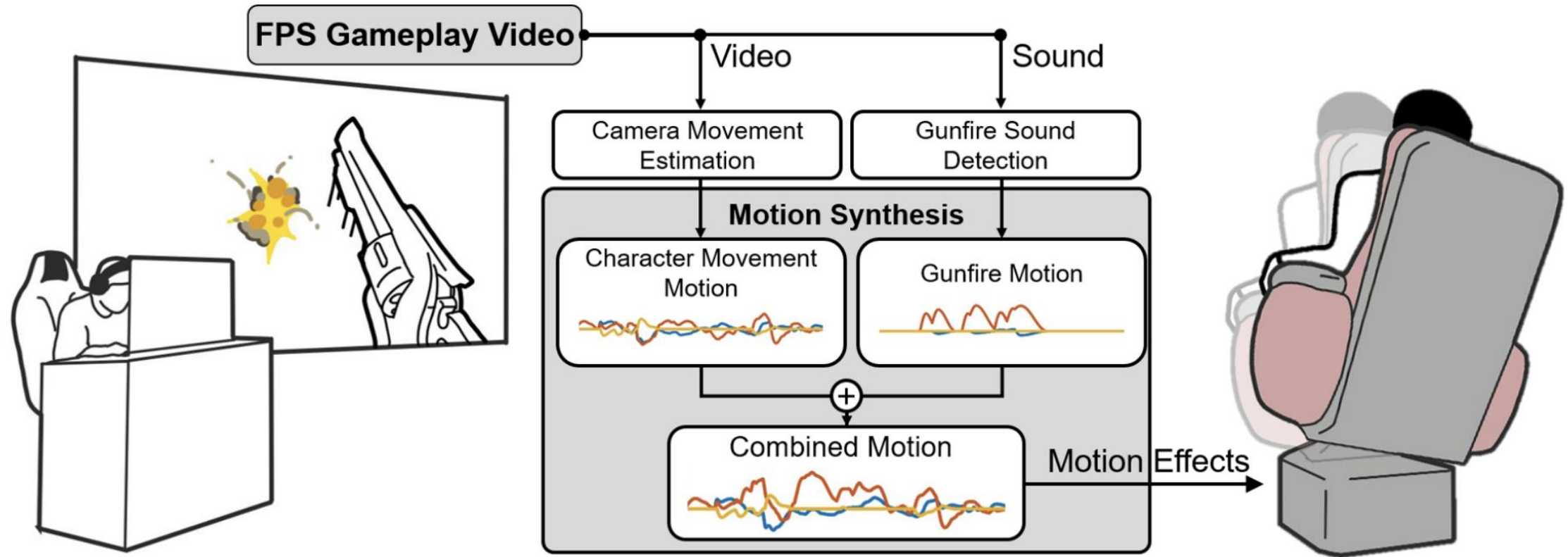
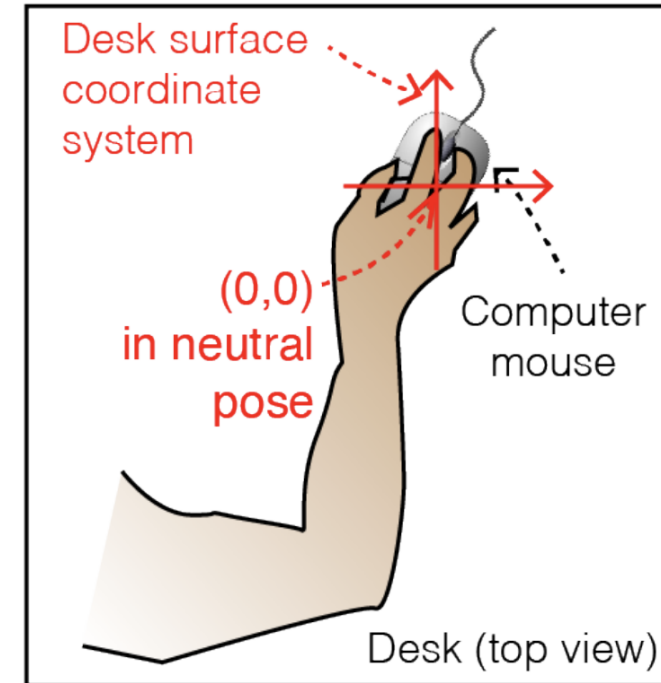
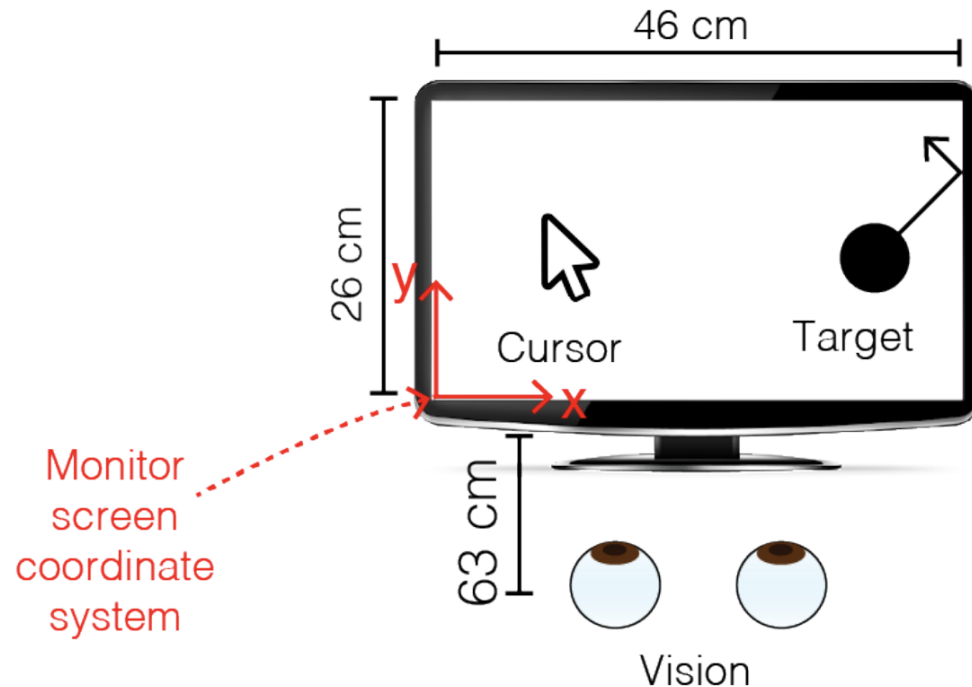


Figure 1: A user watches a play of a first-person shooter game along with automatically generated motion effects. The motion effects emphasize the game character's actions, such as movement and gunfire, for improved viewing experience.

Gyeore Yun, Hyoseung Lee, Sangyoon Han, and Seungmoon Choi. 2021. Improving Viewing Experiences of First-Person Shooter Gameplays with Automatically-Generated Motion Effects. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 320, 1–14. DOI: <https://doi.org/10.1145/3411764.3445358>

A Simulation Model of Intermittently Controlled Point-and-Click Behaviour

Deep Reinforcement Learning



Seungwon Do, Minsuk Chang, and Byungjoo Lee. 2021. A Simulation Model of Intermittently Controlled Point-and-Click Behaviour. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, New York, NY, USA, Article 286, 1–17. DOI: <https://doi.org/10.1145/3411764.3445514>

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<https://dl.acm.org/doi/10.1145/3411764.3445115>
- Vid2Doppler: Synthesizing Doppler Radar Data from Videos for Training Privacy-Preserving Activity Recognition
Radar Data
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